



Definitions and conventions used in organic chemistry

The study of organic chemistry involves recognizing several different types of reactant and reaction. You should find the following summary a useful reference as you work through the chapter.

Types of reactant

Saturated	Unsaturated
- compounds which contain only single bonds - for example: alkanes	- compounds which contain double or triple bonds - for example: alkenes, arenes

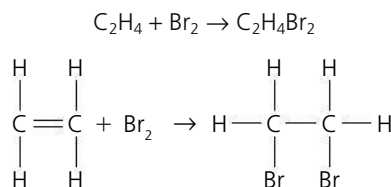
Aliphatics	Arenes
- compounds which do not contain a benzene ring; may be saturated or unsaturated - for example: alkanes, alkenes	- compounds which contain a benzene ring; they are all unsaturated compounds - for example: benzene, phenol

Electrophile (electron-seeking)	Nucleophile (nucleus-seeking)
- an electron-deficient species which is therefore attracted to parts of molecules which are electron rich - electrophiles are positive ions or have a partial positive charge - for example: NO_2^+, H^+, $\text{Br}^{\delta+}$	- an electron-rich species which is therefore attracted to parts of molecules which are electron deficient - nucleophiles have a lone pair of electrons and may also have a negative charge - for example: Cl^-, OH^-, NH_3

Types of reaction

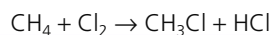
Addition

- occurs when two reactants combine to form a single product
- characteristic of unsaturated compounds
- for example



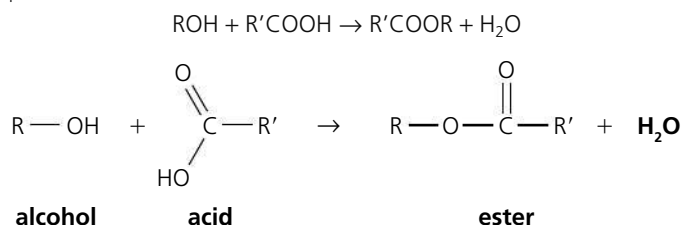
Substitution

- occurs when one atom or group of atoms in a compound is replaced by a different atom or group
- characteristic of saturated compounds and aromatic compounds
- for example



Addition-elimination

- occurs when two reactants join together (addition) and in the process a small molecule such as H_2O , HCl , or NH_3 is lost (elimination)
- reaction occurs between a functional group in each reactant
- also called **condensation** reaction
- for example

**Types of bond breaking (bond fission)**

Homolytic fission	Heterolytic fission
- is when a covalent bond breaks by splitting the shared pair of electrons between the two products - produces two free radicals, each with an unpaired electron $\text{X}:\text{X} \rightarrow \text{X}\cdot + \cdot\text{X}$	- is when a covalent bond breaks with both the shared electrons going to one of the products - produces two oppositely charged ions $\text{X}:\text{X} \rightarrow \text{X}^- + \text{X}^+$

Convention for depicting organic reaction mechanisms

Describing organic reaction mechanisms often involves showing the movement of electrons within bonds and between reactants. The convention adopted for this is a **curly arrow**, drawn from the site of electron availability, such as a pair of non-bonding electrons, to the site of electron deficiency, such as an atom with a partial positive charge.

For example:

$\text{X} \curvearrowright \text{Y}$	represents the electron pair being pulled towards Y so Y becomes δ^- and X becomes δ^+
$\text{X}:\text{---} \rightarrow \text{C}^{\delta+}$	the nucleophile X: is attracted to the electron-deficient C (δ^+)

A 'normal' double-barbed arrow ($\curvearrowright$) represents the motion of an electron pair (as above). Often the mechanism involves several steps. The electrons are transferred ultimately to an atom or group of atoms that then detaches itself and is known as the **leaving group**. We will use blue throughout this chapter to show curly arrows and the pull of electrons.

Note that a single-barbed arrow ($\frown$), known as a **fish-hook**, represents the movement of a single electron. These single arrows are often used in reactions involving radicals.

10.1

Fundamentals of organic chemistry

Understandings:

- A homologous series is a series of compounds of the same family, with the same general formula, which differ from each other by a common structural unit.
- Structural formulas can be represented in full and condensed format.
- Structural isomers are compounds with the same molecular formula but different arrangements of atoms.
- Functional groups are the reactive parts of molecules.
- Saturated compounds contain single bonds only and unsaturated compounds contain double or triple bonds.
- Benzene is an aromatic, unsaturated hydrocarbon.

Guidance

The general formulas (e.g. C_nH_{2n+2}) of alkanes, alkenes, alkynes, ketones, alcohols, aldehydes, and carboxylic acids should be known.

Applications and skills:

- Explanation of the trends in boiling points of members of a homologous series.
- Distinction between empirical, molecular, and structural formulas.
- Identification of different classes: alkenes, alkynes, halogenoalkanes, alcohols, ethers, aldehydes, ketones, esters, carboxylic acids, amines, amides, nitriles, and arenes.
- Identification of typical functional groups in molecules, e.g. phenyl, hydroxyl, carbonyl, carboxyl, carboxamide, aldehyde, ester, ether, amine, nitrile, alkyl, alkenyl, and alkynyl.
- Construction of 3D models (real or virtual) of organic molecules.
- Application of IUPAC rules in the nomenclature of straight-chain and branched-chain isomers.
- Identification of primary, secondary, and tertiary carbon atoms in halogenoalkanes and alcohols and primary, secondary, and tertiary nitrogen atoms in amines.
- Discussion of the structure of benzene using physical and chemical evidence.

Guidance

- Skeletal formulas should be discussed in the course.
- The distinction between class names and functional group names needs to be made, e.g. for OH, hydroxy is the functional group whereas alcohol is the class name.
- The following nomenclature should be covered:
 - non-cyclic alkanes and halogenoalkanes up to haloheptanes
 - alkenes up to hexene and alkynes up to hexyne
 - compounds up to six carbon atoms (in the basic chain for nomenclature purposes) containing only one of the classes of functional groups: alcohols, ethers, aldehydes, halogenoalkanes, ketones, esters, and carboxylic acids.

For a long time it was believed that organic molecules were unique to living things, and could not therefore be synthesized outside a living organism. However, in 1828, the German chemist Friedrich Wöhler synthesized urea from inorganic reactants, commenting in a letter to Berzelius: 'I must tell you that I can make urea without the use of kidneys, either man or dog. Ammonium cyanate is urea'. This discovery, which like many great scientific discoveries was actually made by accident, destroyed the former belief in 'vitalism', and opened the door to the exploration of organic synthesis reactions. The development of new organic compounds is responsible for many of the innovations in our world today.

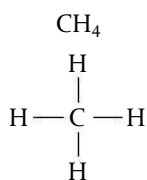
The total number of organic compounds that exists on Earth is so large that it is impossible to estimate with any accuracy. In any case, it is increasing all the time as new materials are synthesized. But we do know that there are at least ten million different organic molecules currently on the planet and every one of them is unique in its chemical structure and specific properties. Of course it is not possible or necessary to study the features of all of the compounds in this large branch of chemistry, but it is useful instead to introduce some system of classification.

Homologous series

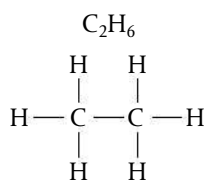
Organic compounds are classified into 'families' of compounds known as **homologous series**. The members of each such series possess certain common features, as described in the following text.

Successive members of a homologous series differ by a $\text{—CH}_2\text{—}$ group

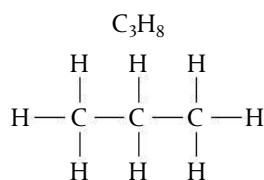
Consider the following compounds of carbon and hydrogen where the carbons are all bonded by single covalent bonds.



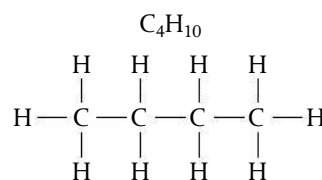
methane



ethane



propane



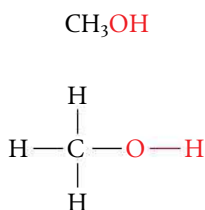
butane

These are members of a homologous series known as the **alkanes**. It can be seen that neighbouring members differ from each other by $\text{—CH}_2\text{—}$. This same increment applies to successive members of each homologous series, and means that the molecular mass increases by a fixed amount as we go up a series.

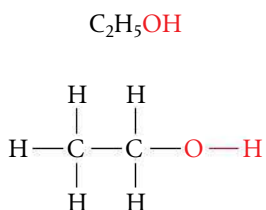
Members of a homologous series can be represented by the same general formula

The four alkanes shown above can all be represented by the general formula $\text{C}_n\text{H}_{2n+2}$.

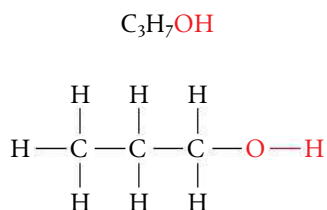
Other homologous series are characterized by the presence of a particular **functional group**, and this will be shown in the general formula for the series. For example, the homologous series known as the **alcohols** possess the functional group —OH as shown below.



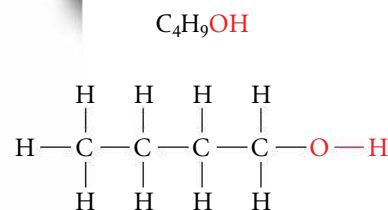
methanol



ethanol



propanol



butanol

These compounds can be represented by the general formula $\text{C}_n\text{H}_{2n+1}\text{OH}$.

The functional group is usually a small group of atoms attached to a carbon atom in a molecule, and it gives the characteristic properties to the compound.

Members of a homologous series show a gradation in physical properties

As successive members of a homologous series differ by a $\text{—CH}_2\text{—}$ group, they have successively longer carbon chains. This is reflected in a gradual trend in the physical properties of the members of the series. For example, the effect of the length of the carbon chain on the boiling point of the alkanes is shown in the table on page 466.

TOK

The word 'organic' is now used in several different contexts beyond the chemistry of carbon compounds. Examples include organic food, organic growth, and organic degeneration. To what extent does this common vocabulary represent a common origin in meaning? Does it confuse or enhance communication?

The trend in boiling points of the alkanes is of great significance in the oil industry, as it makes it possible to separate the many components of crude oil into **fractions** that contain molecules of similar molecular mass on the basis of their boiling points. As the crude oil is heated, the smaller hydrocarbons boil off first while larger molecules distil at progressively higher temperatures in the fractionating column. The different fractions are used as fuels, industrial lubricants, and as starting molecules in the manufacture of synthetic compounds.



Alkane		Boiling point / °C
methane	CH ₄	-164
ethane	C ₂ H ₆	-89
propane	C ₃ H ₈	-42
butane	C ₄ H ₁₀	-0.5
pentane	C ₅ H ₁₂	36
hexane	C ₆ H ₁₄	69
heptane	C ₇ H ₁₆	98
octane	C ₈ H ₁₈	125

gases at room temperature

liquids at room temperature

increasing boiling point

The data show that the boiling point increases with increasing carbon number. This is because of the increased instantaneous induced dipoles causing stronger London (dispersion) forces between the molecules as their molecular size increases. Note, though, that the increase is not linear, but steeper near the beginning as the influence of the increased chain length is proportionally greater for small molecules.

Other physical properties that show this predictable trend with increasing carbon number are density and viscosity.

Oil products. Containers of crude oil and various fractions obtained from it, on a background of a silhouetted oil refinery. The fractions are arranged here in order of increasing boiling point from left to right.

It is estimated that the top ten oil-producing countries of the world produce about two-thirds of the world's oil. This geo-political leverage of the major oil-exporting nations is an on-going challenge to countries that are dependent on imported oil.



Members of a homologous series show similar chemical properties

As they have the same functional group, members of the same homologous series show similar chemical reactivity. For example:

- the alcohols have the —OH functional group, which can be oxidized to form organic acids
- the —COOH functional group, present in the homologous series of the **carboxylic acids**, is responsible for the acidic properties of these compounds.

It follows that if we know the characteristic reactions of a functional group, we will be able to predict the properties of all members of a series.

The main features of a homologous series are as follows:

- Successive members of a homologous series differ by a —CH₂— group.
- Members of a homologous series can be represented by the same general formula.
- The members of a homologous series show a gradation in their physical properties.
- The members of a series have similar chemical properties.

The study of organic chemistry is in some ways like the study of a language where rules have developed to enable communication to be clear and consistent. In order to describe an organic molecule in this way, there are two aspects to consider:

- the formula used to represent the molecule
- the specific name given to the molecule, known as its **nomenclature**.

We will discuss each of these in turn and learn the rules that will enable us to communicate effectively in this study.

TOK

To what extent is chemistry a separate language? What are the main differences between the language of chemistry and your mother tongue?

Formulas for organic compounds: empirical, molecular, and structural

The **empirical formula** of a compound is the simplest whole number ratio of the atoms it contains (see Chapter 1). For example, the empirical formula of ethane, C₂H₆, is CH₃. This formula can be derived from percentage composition data obtained from combustion analysis. It is, however, of rather limited use on its own, as it does not tell us the actual number of atoms in the molecule.

The **molecular formula** of a compound is the actual number of atoms of each element present in a molecule. For example, the molecular formula of ethane is C₂H₆. It is therefore a multiple of the empirical formula, and so can be deduced if we know both the empirical formula and the molar mass *M*. The relationship can be expressed as:

$$M = (\text{molar mass of empirical formula})_n$$

where *n* is an integer.

So, for example, if we know that the empirical formula of ethane is CH₃, and its molar mass *M* = 30 g mol⁻¹, using the formula above:

$$30 \text{ g mol}^{-1} = (\text{mass of CH}_3)_n \text{ so } 30 = (12 + (3 \times 1))_n, \text{ which gives } n = 2$$

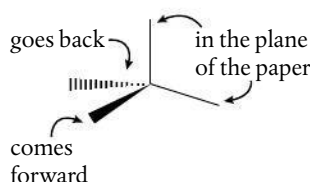
Therefore the molecular formula of ethane is (CH₃)₂ or C₂H₆.

However, the molecular formula is also of quite limited value as the properties of a compound are determined not only by the atoms it contains, but also by how those atoms are arranged in relation to each other and in space.

The **structural formula** is a representation of the molecule showing how the atoms are bonded to each other. There are variations in the amount of detail this shows.



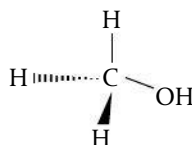
Be careful to mark all the hydrogen atoms with —H when drawing structural formulas. Leaving these out gives what is called a **skeletal structure** and is not acceptable as a structural formula.



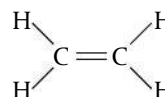
- A **full structural formula** (graphic formula or displayed formula) shows every bond and atom. Usually 90° and 180° (and 120°) angles are used to show the bonds because this is the clearest representation on a two-dimensional page, although in most cases this is not the true geometry of the molecule.
- A **condensed structural formula** often omits bonds where they can be assumed, and groups atoms together. It contains the minimum information needed to describe the molecule non-ambiguously – in other words there is only one possible structure that could be described by this formula.
- A **stereochemical formula** attempts to show the relative positions of atoms and groups around carbon in three dimensions. The convention is that a bond sticking forwards from the page is shown as a solid, enlarging wedge, whereas a bond sticking behind the page is shown as a hashed line. A bond in the plane of the paper is a solid line. When carbon forms four single bonds, the arrangement is tetrahedral with bond angles of 109.5° ; when it forms a double bond, the arrangement is triangular planar with bonds at 120° .

For example:

methanol, CH_3OH



ethene, C_2H_4



The table below shows these different formulas applied to three compounds.

Compound	Ethane	Ethanoic acid	Glucose
Empirical formula	CH_3	CH_2O	CH_2O
Molecular formula	C_2H_6	$\text{C}_2\text{H}_4\text{O}_2$	$\text{C}_6\text{H}_{12}\text{O}_6$
Full structural formula			
Condensed structural formula	CH_3CH_3	CH_3COOH	$\text{CHO}(\text{HCOH})_4\text{CH}_2\text{OH}$

Sometimes we do not need to show the exact details of the hydrocarbon, or **alkyl**, part of the molecule, so we can abbreviate this to **R**. For molecules which contain a benzene ring, C_6H_6 , known as **aromatic compounds**, we use  to show the ring.

Nomenclature for organic compounds: the IUPAC system

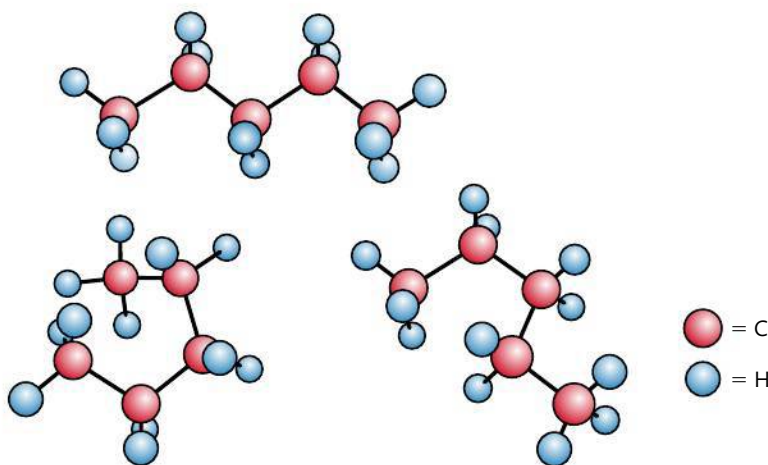
For over a hundred years, chemists have recognized the need for a specific set of rules for the naming of organic compounds. IUPAC names are logically based on the chemistry of the compounds, and so give information about the functional groups present and the size of the molecules. Some guidelines for applying the IUPAC nomenclature are discussed below.

Rule 1: Identify the longest straight chain of carbon atoms

The longest chain of carbon atoms gives the **stem** of the name as follows.

Number of carbon atoms in longest chain	Stem in IUPAC name	Example of compound
1	meth-	CH_4 , methane
2	eth-	C_2H_6 , ethane
3	prop-	C_3H_8 , propane
4	but-	C_4H_{10} , butane
5	pent-	C_5H_{12} , pentane
6	hex-	C_6H_{14} , hexane

Note that 'straight chain' refers to continuous or unbranched chains of carbon atoms – it does not mean angles of 180° . Be careful when identifying the longest straight chain not to be confused by the way the molecule may appear on paper, because of the free rotation around carbon–carbon single bonds. For example, in Figure 10.1, all three structures are the same molecule pentane, C_5H_{12} , even though they may look different.



TOK

Organic chemistry uses many models to represent molecular structures, which differ from each other in the depth of detail. What knowledge do we assume in interpreting these models?



The international, non-governmental organization IUPAC (International Union of Pure and Applied Chemistry), first mentioned in Chapter 1, is best known for its system of nomenclature, now recognized as the world authority in this field. IUPAC terminology enables precise communication between scientists across international boundaries.

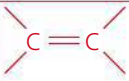
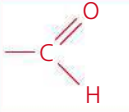
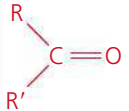
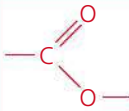
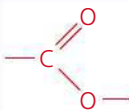
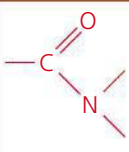
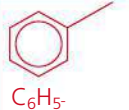


IUPAC nomenclature is seen commonly on labels of everyday substances such as foods, pharmaceuticals, toothpaste, and glue sticks.

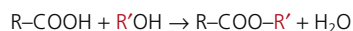
Figure 10.1 Different representations of the same molecule C_5H_{12} , pentane. These can all be interconverted by rotating the carbon–carbon bonds.

Rule 2: Identify the functional group

The functional group usually determines the specific ending or **suffix** to the name, which replaces the '-ane' ending in the parent *alkane*. The suffixes used for some common functional groups are shown in the table below. Note the distinction between *class*, which refers to the type of compound, and *functional group*, which refers to the site of reactivity in the molecule.

Class	Functional group	Name of functional group	Suffix in IUPAC name	Example of compound	General formula
alkane			-ane	C ₂ H ₆ , ethane	C _n H _{2n+2}
alkene		alkenyl	-ene	H ₂ C=CH ₂ , ethene	C _n H _{2n}
alkyne		alkynyl	-yne	HC≡CH, ethyne	C _n H _{2n-2}
alcohol	—OH	hydroxyl	-anol	C ₂ H ₅ OH, ethanol	C _n H _{2n+1} OH
ether	R—O—R'	ether	-oxyalkane	H ₃ C—O—C ₂ H ₅ , methoxyethane	R—O—R'
aldehyde		aldehyde (carbonyl)	-anal	C ₂ H ₅ CHO, propanal	R—CHO
ketone		carbonyl	-anone	CH ₃ COCH ₃ , propanone	R—CO—R'
carboxylic acid		carboxyl	-anoic acid	C ₂ H ₅ COOH, propanoic acid	C _n H _{2n+1} COOH
ester*		ester	-anoate	C ₂ H ₅ COOCH ₃ , methyl propanoate	R—COO—R'
amide		carboxyamide	-anamide	C ₂ H ₅ CONH ₂ , propanamide	
amine	—NH ₂	amine	-anamine	C ₂ H ₅ NH ₂ , ethanamine	
nitrile	—C≡N	nitrile	-anenitrile	C ₂ H ₅ CN, propanenitrile	
arene		phenyl	-benzene	C ₆ H ₅ CH ₃ , methyl benzene	

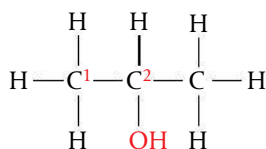
*Esters form when the alkyl group of an alcohol replaces the hydrogen of a carboxylic acid in a condensation reaction:



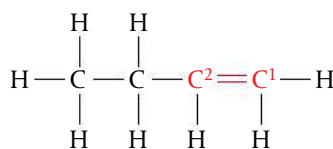
They are named in a similar way to salts, which form when a metal has replaced the hydrogen of a carboxylic acid. Salts take the stem of the name from the parent acid. For example, C₂H₅COONa is **sodium propanoate**. In esters, the alkyl group of the alcohol is the prefix, so C₂H₅COOCH₃ is **methyl propanoate**.

The position of a functional group is shown by a number between dashes inserted before the functional group ending. The number refers to the carbon atom to which the functional group is attached. The chain is numbered starting at the end that will give the smallest number to the group.

For example:



propan-2-ol

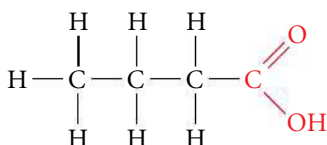


but-1-ene

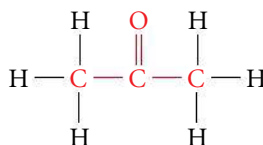
here we number the carbon chain starting from the right-hand side so that the number of the group will be 1 and not 3

Sometimes a functional group can only be in one place, and in these cases we do not need to give a number to show its position.

For example:



butanoic acid



propanone

Rule 3: Identify the side chains or substituent groups

Side chains, or functional groups in addition to the one used as the suffix, are known as **substituents** and are given as the first part or **prefix** of the name. Some common examples are shown in the table.

Class	Functional group	Name of functional group	Prefix in IUPAC name	Example of compound
alkane			methyl, ethyl, propyl, etc.	$\text{CH}_3\text{CH}(\text{CH}_3)\text{C}_2\text{H}_5$, 2-methylbutane $\text{CH}(\text{C}_2\text{H}_5)_3$, 3-ethylpentane $\text{CH}(\text{C}_3\text{H}_7)_3$, 4-propylheptane
halogenoalkane	$-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$	halogeno	fluoro, chloro, bromo, iodo	$\text{C}_2\text{H}_5\text{Cl}$, chloroethane $\text{CH}_3\text{CH}_2\text{BrCH}_3$, 2-bromopropane
amine	$-\text{NH}_2$	amine	amino	$\text{CH}_2(\text{NH}_2)\text{COOH}$, 2-aminoethanoic acid

You will notice that $-\text{NH}_2$ appears as both a possible suffix and a prefix. Usually when it is the only functional group it will take the suffix, but if there are two or more functional groups in the molecule it will be a prefix, as for example in *amino* acids.

As shown above, the position of the substituent groups is given by a number followed by a dash in front of its name showing the carbon atom to which it is attached, again numbering the chain to give the smallest number to the group.



You are expected to be familiar with the nitrogen-containing functional groups at the bottom of the table (amine, amide, and nitrile), but nomenclature of these classes of compounds will not be assessed.

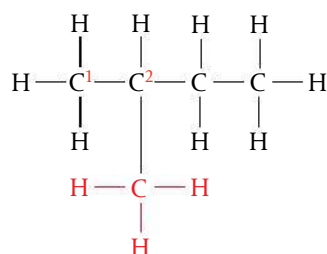


Remember that the name for the stem is derived from the *longest* carbon chain, which may include the carbon of the functional group. Look in the table and see how this applies to carboxylic acids and their derivatives (esters and amides) as well as to nitriles.

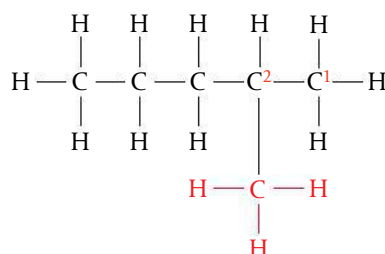


Make sure you don't confuse ethers and esters. Ethers are derived by replacing the H of an alcohol by an alkyl group, esters by replacing the H of an acid by an alkyl group.

For example:



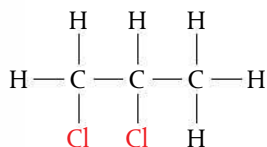
2-methylbutane



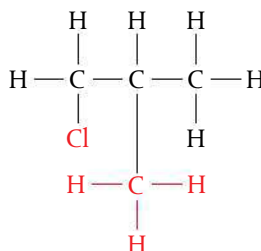
2-methylpentane

If there is more than one substituent group of the same type, we use commas between the numbers and the prefixes di-, tri-, or tetra- before the name. Substituents are given in order of the number of the carbon atom to which they are attached; if there is more than one group on the same atom they are put in alphabetical order.

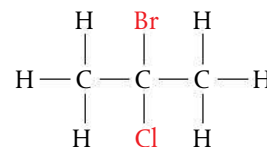
For example:



1,2-dichloropropane

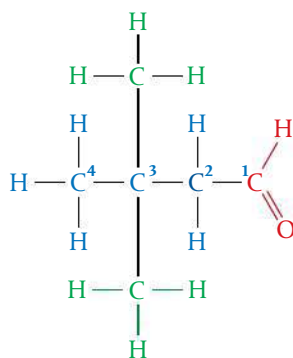


1-chloro-2-methylpropane



2-bromo-2-chloropropane

For example:



3,3-dimethylbutanal

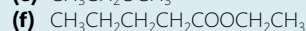
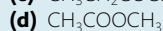
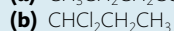
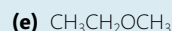
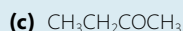
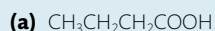
Summary

IUPAC nomenclature has three possible parts, which are usually written together as a single word.

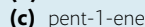
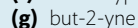
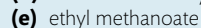
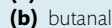
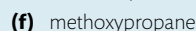
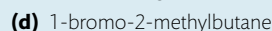
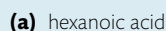
Prefix		Stem		Suffix
position, number and name of substituents		number of carbon atoms in longest chain		class of compound determined by functional group

Exercises

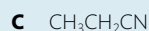
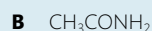
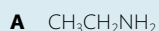
1 For each of the following molecules give its class and its IUPAC name:



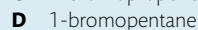
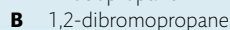
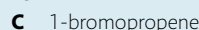
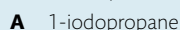
2 Give the structural formulas of the following molecules (condensed form is acceptable):



3 Which of the following is an amine?



4 Which compound is a member of the same homologous series as 1-bromopropane?



There is an H in aldeHyde but not in ketone.

When you are asked to name a compound from a condensed structural formula, it is a good idea to draw out the full structural formula first.

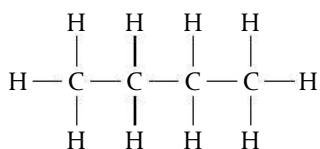
NATURE OF SCIENCE



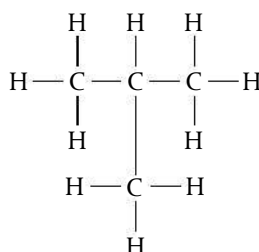
Classification is an important aspect of many disciplines as it brings some order to large amounts of data, and helps in the recognition of patterns and trends. In organic chemistry it is particularly helpful, given the almost infinite number of compounds. The IUPAC nomenclature system described here is broadly accepted by scientists because it provides effective communication. Yet, in common with most systems of classification, there are limits to its usefulness. One of these is in biochemistry, where IUPAC nomenclature can result in seemingly complex and extended names for relatively simple molecules. For example, sucrose, $C_{12}H_{22}O_{11}$, becomes (2*R*,3*R*,4*S*,5*S*,6*R*)-2-[(2*S*,3*S*,4*S*,5*R*)-3,4-dihydroxy-2,5-bis(hydroxymethyl)oxolan-2-yl]oxy-6-(hydroxymethyl)oxane-3,4,5-triol. Although there is a place for such terminology, it is not widely used. Classification and nomenclature are artificial constructs, and when not helpful need not be used. In Chapters 13 and 15 we will commonly use non-IUPAC names for biological molecules for this reason.

Structural isomers: different arrangements of the same atoms

The molecular formula of a compound shows the atoms that are present in the molecule, but gives no information on how they are arranged. Consider for example the formula C_4H_{10} . There are two possible arrangements for these atoms that correspond to different molecules with different properties.

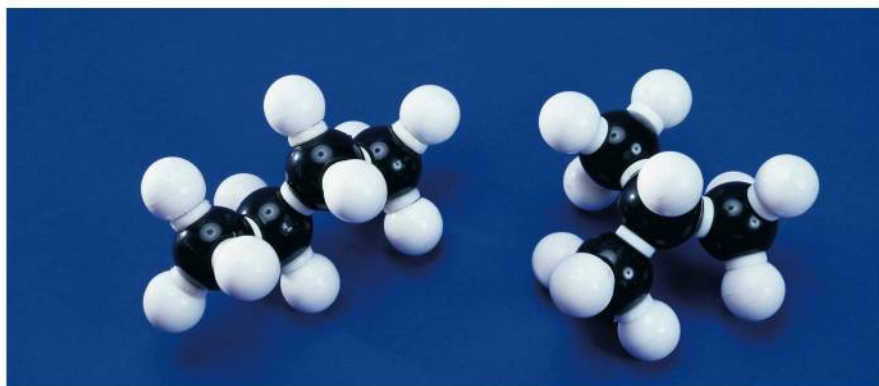


butane,
boiling point -0.5°C



2-methylpropane,
boiling point -11.7°C

Such molecules, having the same molecular formula but different arrangements of the atoms, are known as **structural isomers**.



Each isomer is a distinct compound, having unique physical and chemical properties. As we will see, the number of isomers that exist for a molecular formula increases with the size of the molecule. This is one of the reasons for the vast number of compounds that can be formed from carbon.



Construction of 3D models and working with virtual models online is important experience that helps in gaining an understanding of the structural concepts described here. Full details with worksheets are available online.



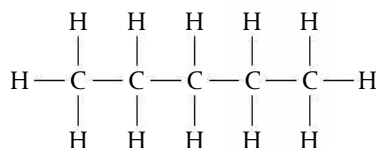
Structural isomers are molecules that have the same molecular formula but different arrangements of the atoms.

Molecular models of the two isomers of C_4H_{10} . On the left is butane (sometimes called *n*-butane to denote a straight chain), and on the right is 2-methylpropane.

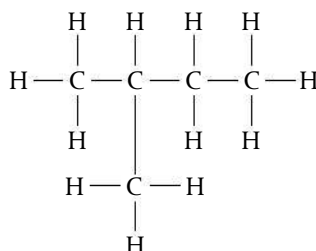
Deducing the structural formulas and names for all possible isomers from a given molecular formula gives us a good opportunity to practise the earlier work on IUPAC nomenclature. We will apply this to isomers in the first six members of the alkanes, the alkenes, and the alkynes.

Structural isomers in alkanes

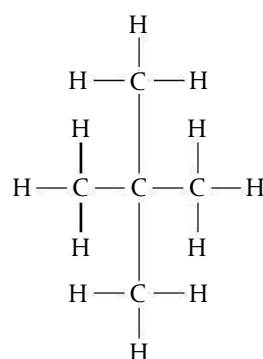
- C_4H_{10} has two isomers, as shown on page 473.
- C_5H_{12} has the following isomers:



pentane

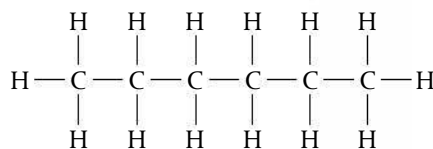


2-methylbutane

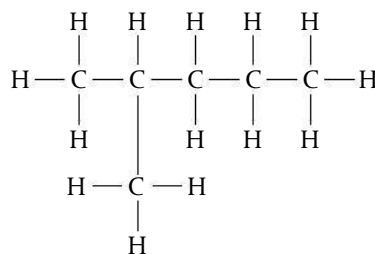


2,2-dimethylpropane

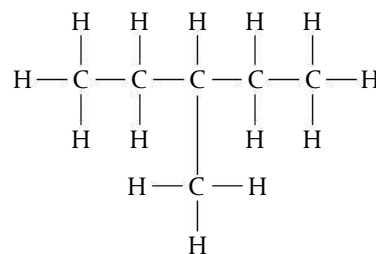
- C_6H_{14} has the following five isomers



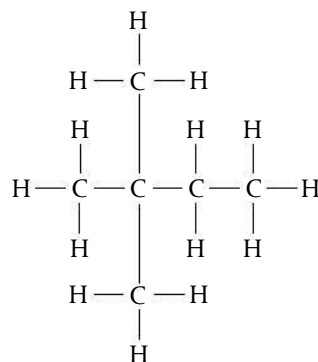
hexane



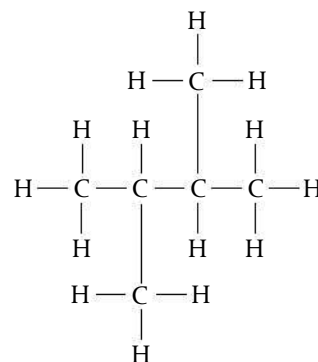
2-methylpentane



3-methylpentane



2,2-dimethylbutane



2,3-dimethylbutane

As we can see here the number of isomers that exist for a molecular formula increases as the molecular size increases. In fact the increase is exponential; there are 75 possible isomers of $C_{10}H_{22}$ and 366 319 of $C_{20}H_{42}$!

When drawing structural formulas, double check that you haven't drawn the same thing twice by drawing it the other way round.

Starting in 1923 a compound called tetraethyl lead – marketed as ‘ethyl’ from America – was added to petroleum in most parts of the world. It proved to be a very successful ‘anti-knocking agent’, allowing lower grades of fuel to be used in internal combustion engines without causing premature burning known as ‘knocking’. Many decades and thousands of tonnes of lead emissions later, mounting concern about the effect of rising lead levels in the atmosphere led to the additive being banned in most countries from about 1986 onwards. Lead is a neurotoxin that is linked to many health effects, particularly brain damage. Since its phase-out from petroleum, blood levels of lead have fallen dramatically but still remain about 600 times higher than they were a century ago.

The person who researched and patented tetraethyl lead as a petroleum additive was the same person who later was responsible for the discovery and marketing of chlorofluorocarbons (CFCs) as refrigerants. Thomas Midgley of Ohio, USA, did not live to know the full extent that the long-term impact his findings would have on the Earth’s atmosphere. He died in 1944, aged 55, from accidental strangulation after becoming entangled in ropes and pulleys he had devised to get himself in and out of bed following loss of use of his legs caused by polio. Perhaps his epitaph should have been ‘The solution becomes the problem’.



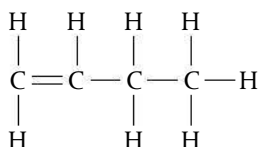
The existence of these different straight-chain and branched-chain isomers of the alkanes is of great significance in the petroleum industry. It has been found that branched-chain isomers generally burn more smoothly in internal combustion engines than straight-chain isomers, and so oil fractions with a higher proportion of these branched-chain isomers are considered to be of ‘better grade’. This is often referred to as a higher ‘octane number’ and means that it fetches a higher price at the pump.



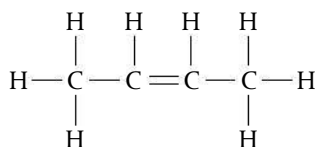
Structural isomers in alkenes

A different type of structural isomer occurs when the carbon–carbon double bond is found in different positions.

- C_4H_8 has the following straight-chain isomers:



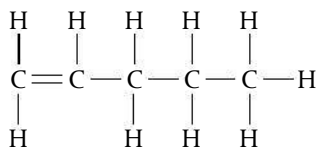
but-1-ene



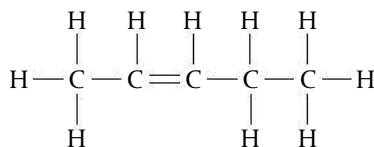
but-2-ene

Note that the molecules are named using the smallest numbered carbon that is part of the double bond.

- C_5H_{10} has the following two straight-chain isomers.

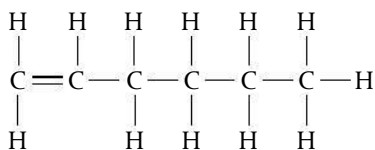


pent-1-ene

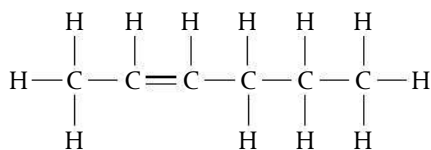


pent-2-ene

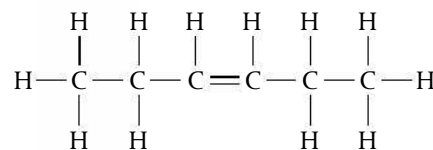
- C_6H_{12} has the following three straight-chain isomers.



hex-1-ene



hex-2-ene



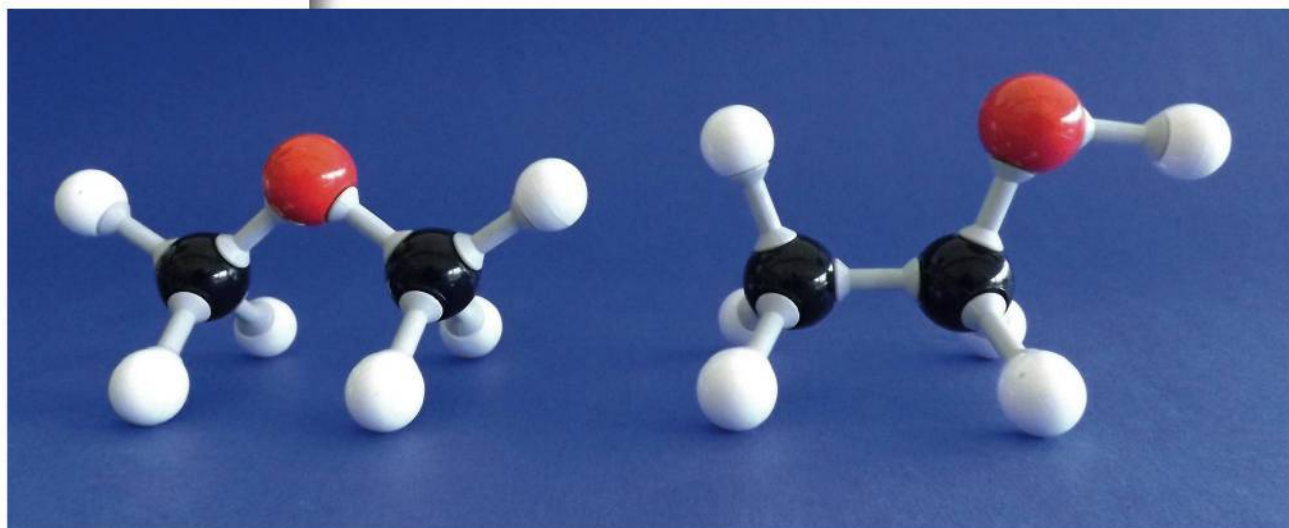
hex-3-ene



A good exercise when practising this work is to give the names that you give to isomers to one of your classmates, and see if they can correctly draw the structures. There should be only one possible structure for each name.

Similarly, the structures and names of the isomers of the alkynes can be deduced from considering different possible positions for the triple bond.

There are many different types of isomers, including those in which the molecules have different functional groups and so are members of different classes of compounds with very different reactivities. For example, the molecular formula C_2H_6O describes both an alcohol and an ether.



Methoxymethane and ethanol are functional group isomers, both with the molecular formula C_2H_6O . Ethanol is widely used in alcoholic drinks while methoxymethane is a gas used in aerosol propellants.

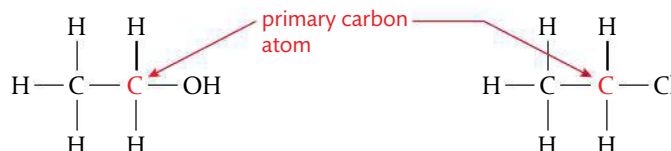
Isomers are of great significance in biochemistry and the drugs industry, as discussed in more detail in Chapters 13 and 15. Later in this chapter, in section 20.3 page 514, we will study a type of isomerism known as stereoisomerism, where the molecules have the same structural formula but different spatial arrangements.

Primary, secondary, and tertiary compounds

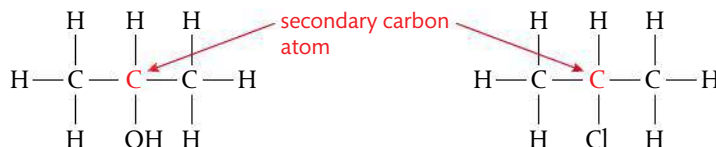
The activity of a functional group is often influenced by its position in the carbon chain, identified as follows.

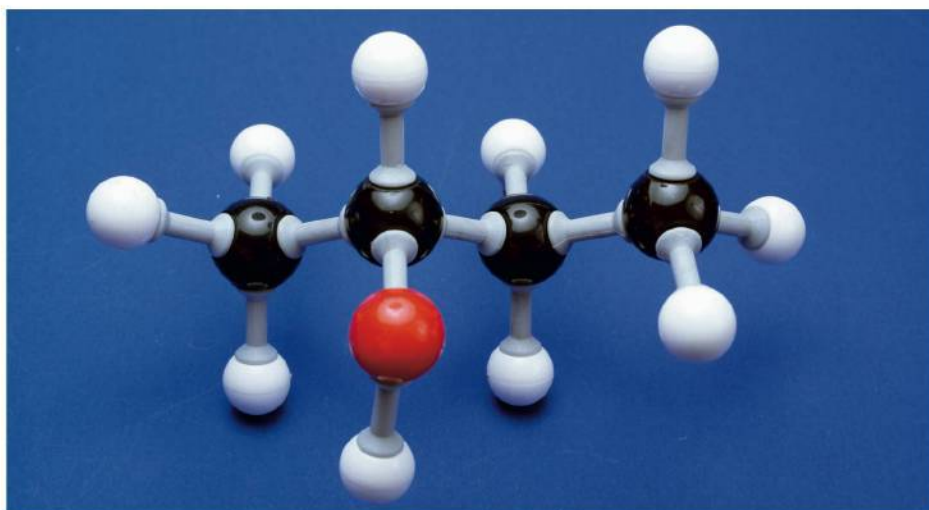
A **primary carbon atom** is attached to the functional group and also to at least two hydrogen atoms. Molecules with this arrangement are known as primary molecules.

For example, ethanol, C_2H_5OH , is a primary alcohol and chloroethane, C_2H_5Cl , is a primary halogenoalkane.



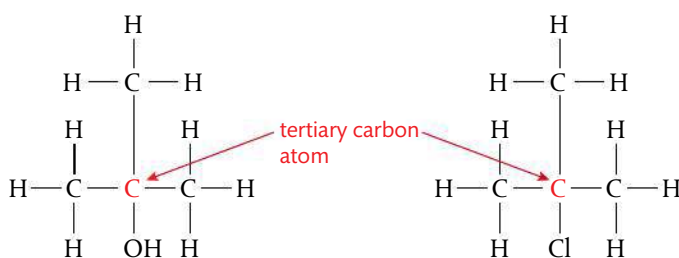
A **secondary carbon atom** is attached to the functional group and also to one hydrogen atom and two alkyl groups. These molecules are known as secondary molecules. For example, propan-2-ol, $CH_3CH(OH)CH_3$, is a secondary alcohol, and 2-chloroethane, $CH_3CHClCH_3$, is a secondary halogenoalkane.



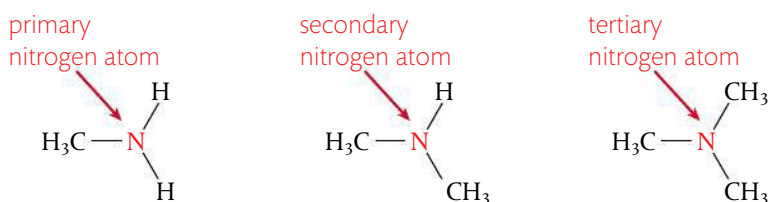


Butan-2-ol, a secondary alcohol. Carbon atoms are black, hydrogen atoms are white, and oxygen atoms are red. Note there is only one hydrogen atom on the carbon attached to the —OH group.

A **tertiary carbon atom** is attached to the functional group and is also bonded to three alkyl groups and so has no hydrogen atoms. These molecules are known as tertiary molecules. For example, 2-methylpropan-2-ol, $\text{C}(\text{CH}_3)_3\text{OH}$, is a tertiary alcohol, and 2-chloro-2-methylpropane, $\text{C}(\text{CH}_3)_3\text{Cl}$, is a tertiary halogenoalkane.



In amines, a similar classification can be applied according to the number of alkyl groups and hydrogen atoms bonded to the nitrogen atom.

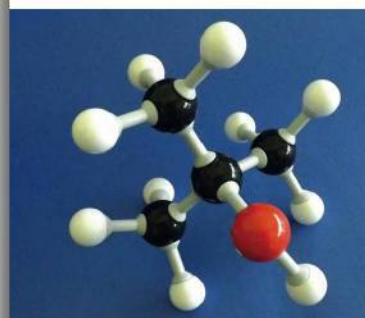


Arenes

Arenes are a class of compounds that are derived from **benzene**, C_6H_6 . They form a special branch of organic compounds known as the **aromatics**, which have properties somewhat distinct from other organic compounds, which are known as **aliphatics**. Arenes contain the phenyl functional group:



The key to understanding the properties of arenes comes from a study of the parent arene molecule – benzene itself.



2-Methylpropan-2-ol, a tertiary alcohol. Carbon atoms are black, hydrogen atoms are white, and oxygen atoms are red. Note there are no hydrogen atoms on the carbon attached to the —OH group.

CHALLENGE YOURSELF

- 1 What is the hybridization of each carbon atom and each nitrogen atom in the tertiary amine drawn here?



Amines and amides, including the substituted derivatives, are widely found in many medicinal compounds. The hormone adrenaline, well known as the 'emergency' hormone as well as a transmitter in the nervous system, is a secondary amine.



Non-polar liquids like benzene can be separated from aqueous solutions using a separating funnel because they form a separate layer and the lower liquid can be drained. Here, the lower aqueous layer has been coloured purple to make the separation more visible.

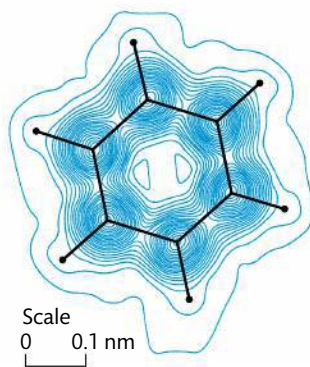
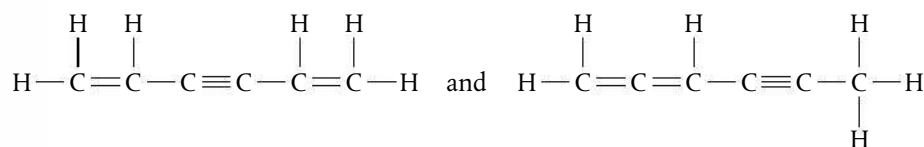


Figure 10.2 Electron density map for benzene at -3°C . The blue lines join parts of the molecule with equal electron density, showing that all bonds between carbon atoms are equal and of length 0.139 nm.

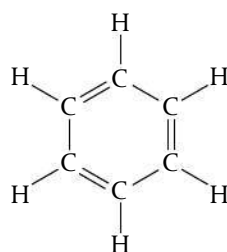
Benzene does not behave like other unsaturated molecules

The 1 : 1 ratio of carbon to hydrogen in benzene, C_6H_6 , indicates a high degree of unsaturation, greater than that of alkenes or alkynes. Yet early observations on benzene indicated that it does not show the characteristic properties of possible structures such as:



For example, benzene has no isomers, and is reluctant to undergo addition reactions – both features that would be expected from these molecules.

In 1865 Kekulé suggested a cyclic arrangement of the carbon atoms with alternating single and double bonds. This is 1,3,5-cyclohexatriene:



which is usually drawn without the atoms marked as



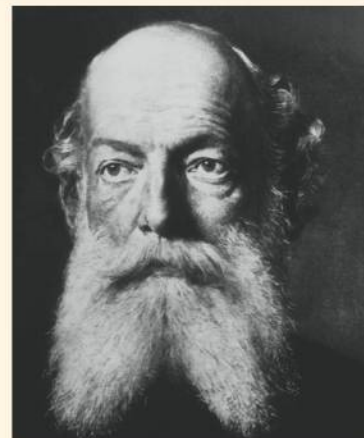
The symmetry of this model explained some of the known properties of benzene, such as the fact that it has no isomers, but again fell short of explaining benzene's atypical reactivity for such an unsaturated molecule. Clearly, modifications to the Kekulé structure were needed. Advances in technology, such as X-ray diffraction, provided data that helped to develop the current model of benzene structure.



Benzene is an important industrial solvent and is used in the synthesis of drugs, dyes, and plastics such as polystyrene. Because of its frequent use, benzene has become widespread in the environment of developed countries. In the USA, petroleum contains up to 2% benzene by volume and it may be as high as 5% in other countries. But benzene is a toxic compound and a known carcinogen. Chronic exposure at low levels can lead to aplastic anaemia and leukaemia. Regulating bodies in many countries have set permissible exposure limits for benzene levels in drinking water, foods, and air in the workplace.



Friedrich August Kekulé von Stradonitz (1829–1896) was a German organic chemist who also spent parts of his working life in France, Switzerland, and England. He was a principal founder of the theory of chemical structure, establishing the tetravalence of carbon and the ability of carbon atoms to link to each other. He provided the first molecular formulas using lines to represent bonds between atoms, and most famously the ring structure of benzene. In addition, he was evidently an influential teacher as three of the first five Nobel Prizes in Chemistry were won by his students: van't Hoff in 1901, Fischer in 1902, and Baeyer in 1905.



The special stability of the benzene ring is the result of delocalized electrons

Benzene is a cyclic structure, in which a framework of single bonds attaches each carbon to the one on either side and to a hydrogen atom. Each of the six carbon atoms is sp^2 hybridized, and forms three sigma σ bonds with angles of 120° , making a planar shape.

This leaves one unhybridized p electron on each carbon atom with its dumb-bell shape perpendicular to the plane of the ring. But instead of pairing up to form discrete alternating π bonds, the p orbitals effectively overlap in both directions, spreading themselves out evenly to be shared by all six carbon atoms. This forms a **delocalized π electron cloud** in which electron density is concentrated in two donut-shaped rings above and below the plane of the ring. This is a very stable arrangement and lowers the internal energy of the molecule.

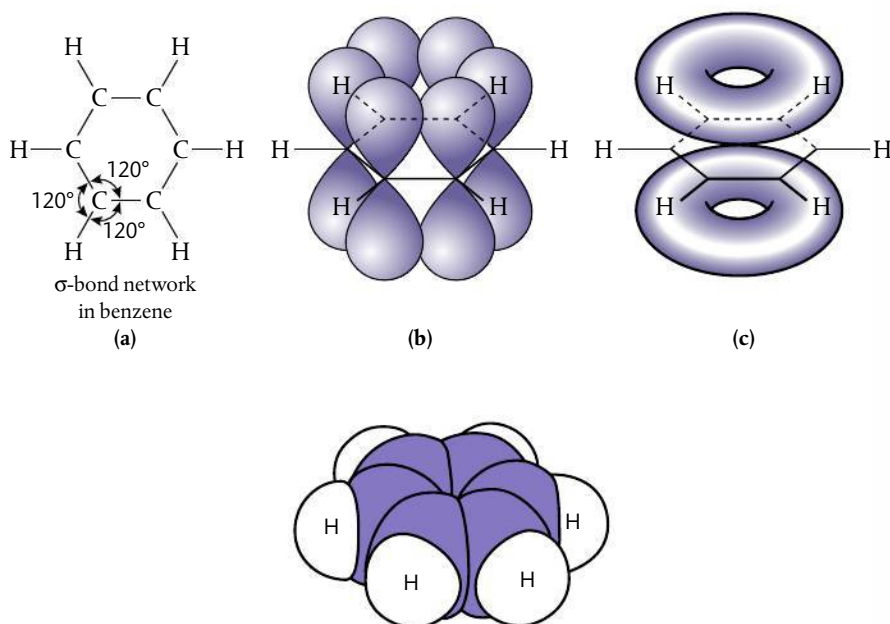
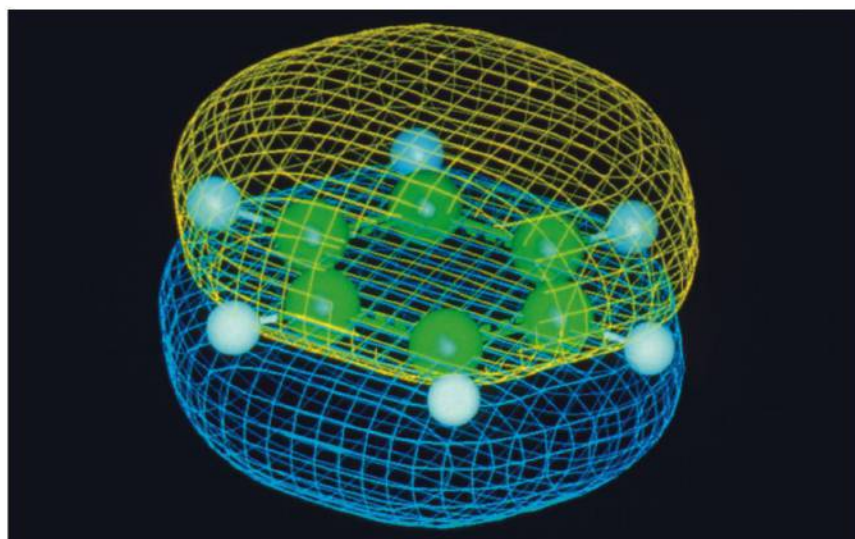


Figure 10.3 The bonding in benzene. This is a symmetrical arrangement where all the p electrons are shared equally by all of the bonded carbon atoms. It is a very stable structure and lowers the internal energy of the molecule. **(a)** The σ -bond framework. **(b)** The unhybridized p orbitals before overlapping sideways. **(c)** The delocalized π electrons form regions of high electron density above and below the planar benzene ring.

Figure 10.4 A space-filling model of benzene, showing its planar shape.



Computer graphics representation of the structure of benzene showing the delocalized electron clouds which confer great stability on the molecule. They appear as the blue and yellow cages above the flat plane of the molecule. Carbon atoms are shown in green and hydrogen atoms in white.

TOK

There is a well-established account that Kekulé visualized the ring structure of benzene after experiencing a dream about snakes biting each other's tails. To what extent can imagination and unconscious processes be important in scientific advancement?

Make sure that when you are writing the structure of benzene you do not forget to include the ring inside the hexagon. If you do forget, and draw instead



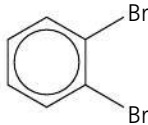
, this denotes an entirely different structure, cyclohexane C_6H_{12} .



The usual convention for depicting this structure of benzene uses a ring inside the hexagon to denote the delocalized electrons.



This model is supported by evidence, and provides explanations for most observed properties of benzene, as described below.

Property	Observation / evidence	Explanation
1 Bond lengths	All carbon–carbon bond lengths in benzene are equal and intermediate in length between single and double bonds.	Each bond contains a share of three electrons between the bonded atoms. X-ray bond length data / nm: alkane single bond $C—C$ 0.154 nm alkene double bond $C=C$ 0.134 nm benzene  0.139 nm
2 $\Delta H_{\text{hydrogenation}}$ for the reaction $C_6H_6 + 3H_2 \rightarrow C_6H_{12}$	Theoretical value based on adding H_2 across three $C=C$ double bonds = -362 kJ mol^{-1} Experimental value for benzene = -210 kJ mol^{-1} Benzene is more stable than predicted from the Kekulé structure by approx. 152 kJ mol^{-1}	Delocalization minimizes the repulsion between electrons and so gives benzene a more stable structure, lowering its internal energy by 152 kJ mol^{-1} . This is the amount of energy that would have to be supplied in order to overcome the special stability of the delocalized ring. It is called the resonance energy or the stabilization energy of benzene. See Figure 10.5.
3 Type of reactivity	Benzene is reluctant to undergo addition reactions and is more likely to undergo substitution reactions.	Addition reactions are energetically not favoured as they would involve disrupting the entire cloud of delocalized electrons. The resonance energy would have to be supplied and the product, without the delocalized ring of electrons, would be <i>less</i> stable. Benzene can instead undergo substitution reactions that preserve the stable ring structure.
4 Isomers	Only one isomer exists of compounds such as 1,2-dibromobenzene.	As benzene is a symmetrical molecule with no alternating single and double bonds, all adjacent positions in the ring are equal. For example 1,2-dibromobenzene, $C_6H_4Br_2$ 

The reactivity of benzene is discussed further on page 508.

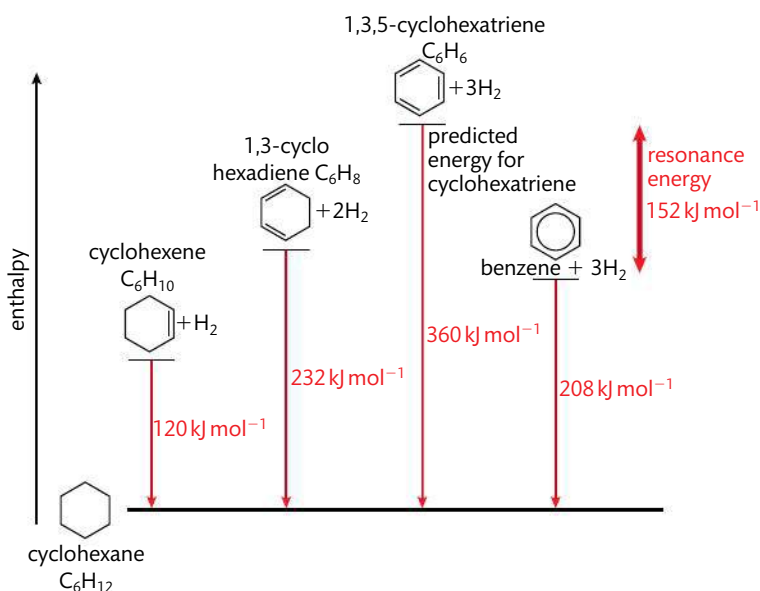


Figure 10.5 Enthalpy changes for the hydrogenation of benzene and related molecules. Note that the molecule 1,3,5-cyclohexatriene does not exist, so the value for its hydrogenation is theoretical only.



NATURE OF SCIENCE

The changing model of the structure of benzene has largely been a result of advances in technology and instrumentation. The Kekulé model seemed to explain many of the known properties of benzene at the time, but had to be reconsidered when quantitative data obtained through X-ray diffraction experimentation provided counter evidence, for example on bond lengths. The progress of science sometimes demands a refinement or abandonment of a model as new data is obtained, and the acceptance of this is dependent on the objectivity and openness of scientists.

Trends in physical properties

We have seen that the structure of organic compounds can be thought of in two parts.

- 1 A framework consisting of carbon and hydrogen only, known as the **hydrocarbon skeleton**. This differs in size in different members of the same homologous series.
- 2 A functional group. This differs in identity in different homologous series.

Both of these components influence the physical properties of a compound. Most of these properties were discussed in relation to intermolecular forces in Chapter 4, page 178, and a few additional points are made here.

The increase in boiling point with carbon number that we saw in the alkanes applies to other homologous series. In general, the lower members of a series are likely to be gases or liquids at room temperature, while the higher members are more likely to be solids. Branching of the hydrocarbon chain also has an effect on volatility as it influences the strength of the intermolecular forces. Think, for example, how tree logs with lots of branches sticking out cannot stack together as closely as a pile of logs which have no branches. In a similar way branched-chain isomers have less contact with each other than their straight-chain isomers, and so have weaker intermolecular forces and so lower boiling points. We saw earlier that the straight-chain molecule butane has a higher boiling point (-0.5°C) than its branched-chain isomer, 2-methylpropane (boiling point -11.7°C).

Owing to their different polarities and resulting differences in intermolecular forces, functional groups have a significant effect on volatility. This can be summarized as follows.

most volatile

least volatile

alkane > halogenoalkane > aldehyde > ketone > alcohol > carboxylic acid

London (dispersion) force $\rightarrow$ dipole-dipole interaction $\rightarrow$ hydrogen bonding

increasing strength of intermolecular attraction $\longrightarrow$

increasing boiling point $\longrightarrow$

Exercises

- 5 Draw and name all the structural isomers of $C_3H_3Cl_5$.
- 6 Which formula is that of a secondary halogenoalkane?
A $CH_3CH_2CH_2CH_2Br$ **B** $CH_3CHBrCH_2CH_3$ **C** $(CH_3)_2CHCH_2Br$ **D** $(CH_3)_3CBr$
- 7 Describe the bonding in a benzene molecule and use it to explain benzene's energetic stability.
- 8 (a) When comparing the boiling points of different classes of compound, why is it important to choose molecules that have similar molar mass?
 (b) Explain how you would expect the solubility of alcohols in hexane to change with increasing chain length.

10.2 Functional group chemistry

Understandings:

- **Alkanes:** have low reactivity and undergo free radical substitution reactions.
- **Alkenes:** are more reactive than alkanes and undergo addition reactions. Bromine water can be used to distinguish between alkenes and alkanes.
- **Alcohols:** undergo nucleophilic substitution reactions with acids (also called esterification or condensation) and some undergo oxidation reactions.
- **Halogenoalkanes:** are more reactive than alkanes. They can undergo (nucleophilic) substitution reactions. A nucleophile is an electron-rich species containing a lone pair that it donates to an electron-deficient carbon.
- **Polymers:** addition polymers consist of a wide range of monomers and form the basis of the plastics industry.
- **Benzene:** does not readily undergo addition reactions but does undergo electrophilic substitution reactions.

Applications and skills:

- **Alkanes:**
 - Writing equations for the complete and incomplete combustion of hydrocarbons.
 - Explanation of the reaction of methane and ethane with halogens in terms of a free radical substitution mechanism involving photochemical homolytic fission.
- **Alkenes:**
 - Writing equations for the reactions of alkenes with hydrogen and halogens and of symmetrical alkenes with hydrogen halides and water.
 - Outline of the addition polymerization of alkenes.
 - Relationship between the structure of the monomer to the polymer and repeating unit.
- **Alcohols:**
 - Writing equations for the complete combustion of alcohols.
 - Writing equations for the oxidation reactions of primary and secondary alcohols (using acidified potassium dichromate(VI) or potassium manganate(VII) as oxidizing agents). Explanation of distillation and reflux in the isolation of the aldehyde and carboxylic acid products.
 - Writing the equation for the condensation reaction of an alcohol with a carboxylic acid, in the presence of a catalyst (e.g. concentrated sulfuric acid) to form an ester.
- **Halogenoalkanes:**
 - Writing the equation for the substitution reactions of halogenoalkanes with aqueous sodium hydroxide.

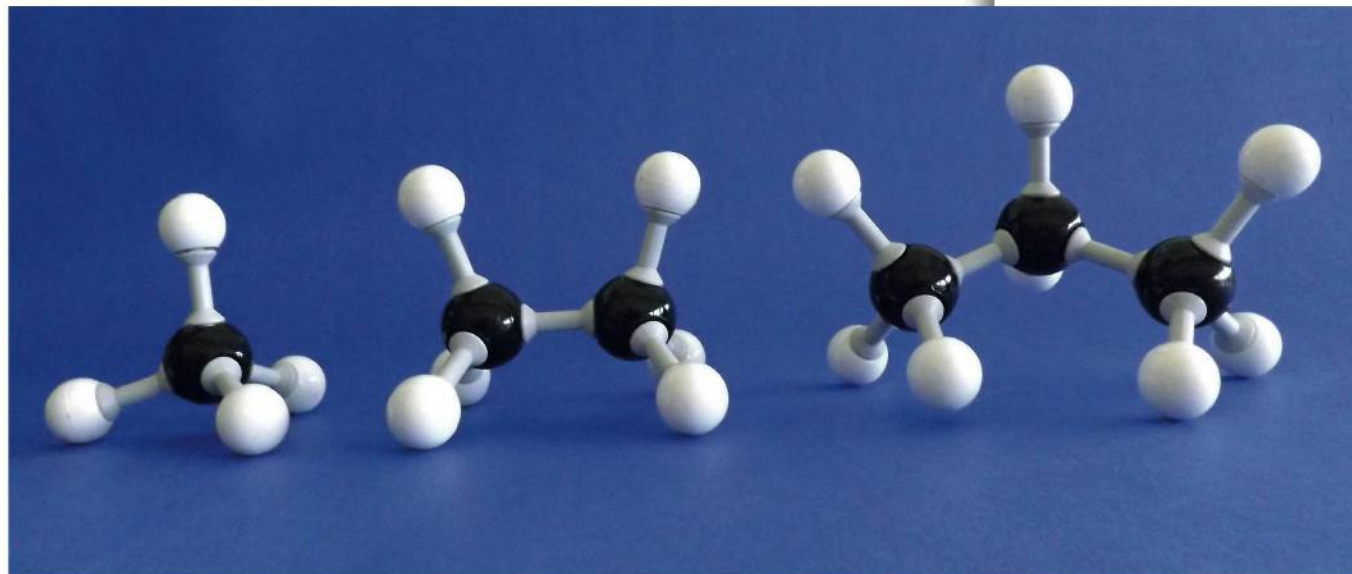
Guidance

- Reference should be made to initiation, propagation, and termination steps in free radical substitution reactions. Free radicals should be represented by a single dot.
- The mechanisms of S_N1 and S_N2 and electrophilic substitution reactions are not required.

When asked to define a hydrocarbon, make sure you include the word *only* after the fact that they contain carbon and hydrogen – as otherwise your description would apply to *all* organic compounds!



Alkanes



- General formula is C_nH_{2n+2}
- Alkanes are **saturated hydrocarbons**.

The term hydrocarbons refers to compounds which contain carbon and hydrogen *only*. Alkanes are said to be saturated because they contain all single carbon–carbon bonds.

Alkanes contain only C—C and C—H bonds, which are both strong bonds: C—C = 348 kJ mol^{-1} and C—H = 412 kJ mol^{-1} . So these molecules will only react in the presence of a strong source of energy, strong enough to break these bonds. As a result, alkanes are stable under most conditions and can be stored, transported, and even compressed safely – which is one of the reasons why they are such useful compounds.

The C—C and C—H bonds are also characteristically non-polar, so these molecules are not susceptible to attack by most common reactants. These two factors taken together mean that alkanes are generally of very low reactivity.

There are, however, two reactions involving alkanes that we will consider here – combustion and substitution reactions.

Combustion: alkanes as fuels

Alkanes are widely used as fuels, in internal combustion engines and household heating for example, because they release significant amounts of energy when they burn. In other words, the reactions of combustion are highly exothermic. This is mainly because of the large amount of energy released in forming the double bonds in CO_2 and the bonds in H_2O .

Alkanes burn in the presence of excess oxygen to produce carbon dioxide and water. This is known as complete combustion because the products are fully oxidized.



Molecular models of the first three members of the alkane homologous series: methane (CH_4), ethane (C_2H_6), and propane (C_3H_8). Carbon atoms are shown in black and hydrogen atoms in white. There is free rotation around the carbon–carbon bonds in these saturated molecules.



Methane, CH_4 , is both a fuel and an atmospheric pollutant. As a greenhouse gas, it has a much higher potential for trapping infrared radiation than carbon dioxide. It is emitted from natural sources such as wetlands, but about 60% of CH_4 released is from human activities. The three main sources are the petroleum industry, domestic livestock, and landfill sites. Strategies to reduce CH_4 emission include its capture and use as a source of energy.

Aeroplane landing with exhaust behind. The burning of hydrocarbon fuels releases large amounts of carbon dioxide, as well as carbon monoxide and other pollutants, into the atmosphere.



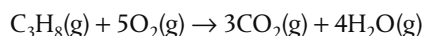
Given all the negative environmental effects associated with the burning of fossil fuels, how can we make changes in our lifestyles to lessen these impacts?



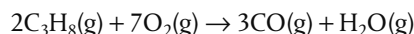
CHALLENGE YOURSELF

- 2 Use oxidation numbers to show the differences in the extent of oxidation in reactions of complete and incomplete combustion of hydrocarbons.

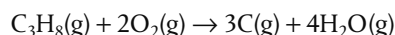
For example:



However, when the oxygen supply is limited, incomplete combustion occurs giving rise to carbon monoxide and water. For example:



In conditions of extreme oxygen limitation, carbon will also be produced. For example:



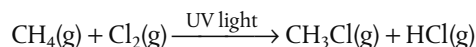
Other hydrocarbons, the alkenes, alkynes, and arenes, similarly undergo complete or incomplete combustion depending on the availability of oxygen, with the release of large amounts of energy. As the C:H ratio increases with unsaturation, there is an increase in the smokiness of the flame, due to unburned carbon.

The products of all these reactions have serious impact on the environment, which is why the burning of these and other **fossil fuels** on a very large scale is widely recognized as a global problem.

- Carbon dioxide and water are both so-called **greenhouse gases**, which means that they absorb infrared radiation and so contribute to **global warming** and **climate change**. Rising levels of carbon dioxide are largely implicated in the increase in average world temperatures over the last century.
- Carbon monoxide is a toxin because it combines irreversibly with the hemoglobin in the blood and prevents it from carrying oxygen. Slow idling vehicle engines in regions of high traffic densities produce higher concentrations of carbon monoxide. It is essential to provide adequate ventilation when these fuels are being burned in a confined space, to avoid carbon monoxide poisoning which can be fatal.
- Unburned carbon is released into the air as particulates, which have a direct effect on human health, especially the respiratory system. In addition, these particulates act as catalysts in forming smog in polluted air, and have also been targeted as the source of another serious environmental problem known as **global dimming**.

Substitution reactions of alkanes: halogenation

As alkanes are saturated molecules, the main type of reaction that they can undergo is **substitution**. This occurs when another reactant, such as a halogen, takes the place of a hydrogen atom in the alkane. For example, methane, CH_4 , reacts with chlorine producing chloromethane and hydrogen chloride.



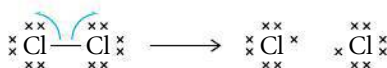
The reaction does not take place in the dark as the energy of UV light is necessary to break the covalent bond in the chlorine molecule. This splits it into chlorine atoms, which each have an unpaired electron and are known as **free radicals**. Once formed, these radicals will start a chain reaction in which a mixture of products including the halogenoalkane is formed. We can describe the reaction as a sequence of steps, known as the **reaction mechanism**.

Initiation



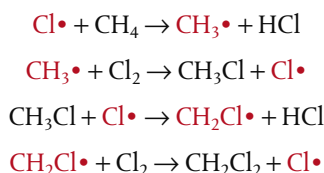
This process occurs in the presence of ultra-violet light. It is known as **photochemical homolytic fission** because the bond between the chlorine atoms is broken, splitting the shared pair of electrons between the two atoms. *Homo-* means 'the same' and refers to the fact that the two products have an equal assignment of electrons from the bond.

This process can be shown using single-sided curly arrows, sometimes known as 'fish-hooks', to show the movement of a single electron.



Propagation

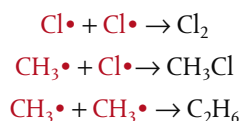
Propagation reactions both use and produce free radicals. For example:



There are many possible propagation steps, which all allow the reaction to continue. This is why this type of reaction is often called a **chain reaction**.

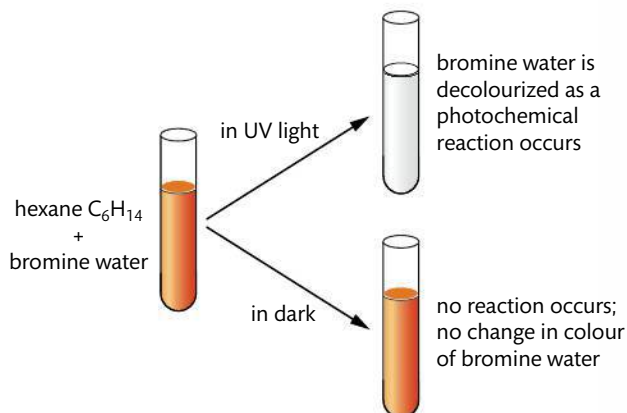
Termination

Termination reactions remove free radicals from the mixture by causing them to react together and pair up their electrons. For example:



Again there are many possible termination steps.

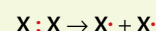
So the reaction mixture may contain mono- and di-substituted halogenoalkanes, as well as HCl and larger alkanes. A similar reaction occurs with other alkanes and with bromine. The change in colour from brown to colourless of bromine water when it reacts with an alkane occurs only in the presence of UV light, as shown in Figure 10.6. This reaction is sometimes used to distinguish between alkanes and alkenes.



A free radical contains an unpaired electron and so is very reactive.



Homolytic fission occurs when a bond breaks by splitting the shared pair of electrons between the two products. It produces two free radicals, each with an unpaired electron:



Make sure that you understand the difference between a free radical and an ion. A free radical has an unpaired electron but no net charge; an ion carries a charge.

Figure 10.6 The reaction between hexane, C_6H_{14} , and bromine water takes place only in the presence of UV light.



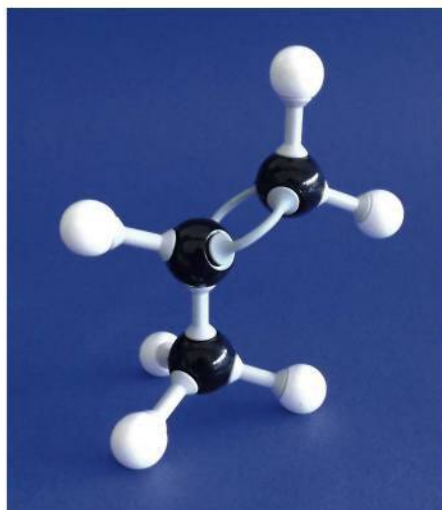
Alkanes are saturated hydrocarbons and undergo substitution reactions.

A molecular model of propene, C_3H_6 , showing the carbon-carbon double bond between two of the carbon atoms. The double bond is the site of reactivity of the alkenes, as the π bond can break enabling the molecule to undergo addition reactions. This is why the alkenes are described as being unsaturated.

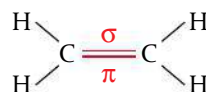
CHALLENGE YOURSELF

- 3 The mechanism for the addition reactions of alkenes is completely different from the substitution reactions we discussed for alkanes; addition reactions involve **heterolytic fission**. Think about what this term suggests about the type of bond breaking involved. What might the mechanism for the addition reaction be, based on this? This is discussed later in this chapter.

Alkenes



- General formula C_nH_{2n} .
- Alkenes are **unsaturated hydrocarbons** containing a carbon-carbon double bond.

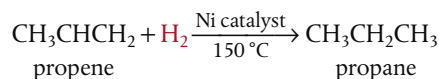
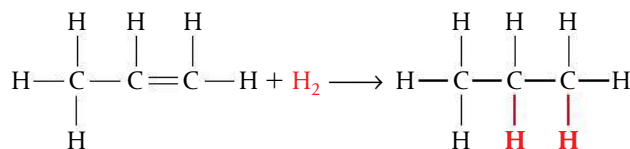


As we learned in Chapter 4, the double bond is made of two different bonds, one sigma, σ , and one pi, π . The carbon atoms are sp^2 hybridized, forming a trigonal planar shape with angles of 120° .

Alkenes are more reactive than alkanes as the double bond is the site of reactivity of the molecule. The π bond is relatively easily broken, which creates two new bonding positions on the carbon atoms. This enables alkenes to undergo **addition reactions** and so form a range of different saturated products. These are described in the following text.

Addition of hydrogen

Hydrogen reacts with alkenes to form alkanes in the presence of a nickel catalyst at about 150°C .



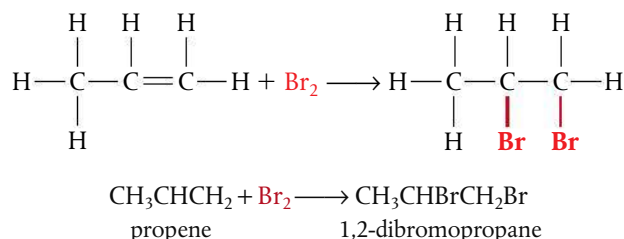
This process, known as **hydrogenation**, is used in the margarine industry to convert oils containing many unsaturated hydrocarbon chains into more saturated compounds which have higher melting points. This is done so that margarine will be a solid at room temperature. However, there are now widespread concerns about the health effects of so-called **trans fats**, produced by partial hydrogenation. This is discussed further in Chapter 13, Biochemistry.



Oct-1-ene, C_8H_{16} , an alkene, burning in a crucible. The flame is smoky due to unburned carbon, an indication of unsaturation.

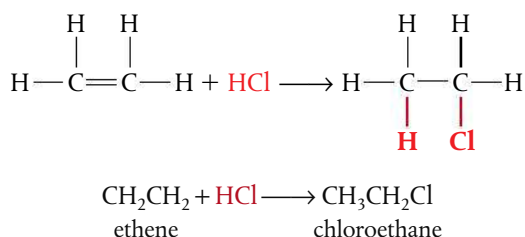
Addition of halogens

Halogens react with alkenes to produce dihalogeno compounds. These reactions happen quickly at room temperature, and are accompanied by the loss of colour of the reacting halogen. Note that the name and structure of the product indicates that a halogen atom attaches to each of the two carbon atoms of the double bond.



Addition of halogen halides

Hydrogen halides, such as HCl and HBr, react with alkenes to produce halogenoalkanes. These reactions take place rapidly in solution at room temperature.

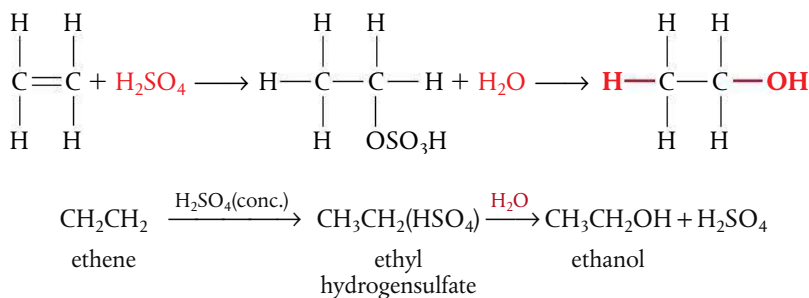


All the hydrogen halides are able to react in this way, but the reactivity is in the order $\text{HI} > \text{HBr} > \text{HCl}$ due to the decreasing strength of the hydrogen halide bond down Group 17. So HI, with the weakest bond, reacts the most readily.

The mechanism of these reactions is discussed in Section 20.1.

Addition of water

The reaction with water is known as **hydration**, and converts the alkene into an alcohol. In the laboratory, it can be achieved using concentrated sulfuric acid as a catalyst. The reaction involves an intermediate in which both H^+ and HSO_4^- ions are added across the double bond, but this is quickly followed by hydrolysis with replacement of the HSO_4^- by OH^- and reformation of the H_2SO_4 .



Conditions: heat with steam and catalyst of concentrated H_2SO_4

This reaction is of industrial significance because ethanol is a very important solvent, and so is manufactured on a large scale.



Alkenes are unsaturated hydrocarbons and undergo addition reactions.



Be careful not to confuse the terms *hydrogenation* (addition of hydrogen) with *hydration* (addition of water).



The modern process for ethanol synthesis is by direct catalytic hydration of ethene over a phosphoric acid catalyst absorbed on silica.



NATURE OF SCIENCE

Two observable differences in the reactions of alkanes and alkenes have been mentioned in this section.

- Alkenes decolorize bromine water rapidly at room temperature, whereas alkanes will only do this in the presence of UV light.
- Alkenes burn with a dirtier, smokier flame than alkanes.

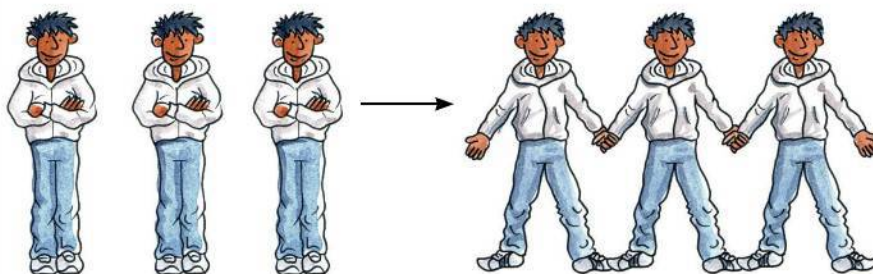
Either of these tests might be considered as a way to distinguish between the two classes of compound, but both rely on qualitative data and are subjective, based on the eye of the observer. Wherever possible, experiments are designed to produce quantitative data that is more objective and able to be analysed. Consider what instrumentation and sensors could be used to develop these observations into more rigorous tests to distinguish between saturated and unsaturated hydrocarbons.

Use of bromine water to distinguish between an alkane (hexane) and an alkene (hex-1-ene). Without UV light, the brown colour is decolorized by the alkene but not by the alkane. This is due to the addition reaction that occurs with the unsaturated alkene that does not occur with the alkane.

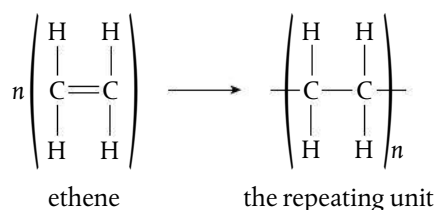
Polymerization of alkenes

Because alkenes readily undergo addition reactions by breaking their double bond, they can be joined together to produce long chains known as **addition polymers**. The alkene used in this reaction is known as the **monomer**, and its chemical nature determines the properties of the polymer. Polymers, typically containing thousands of molecules of the monomer, are a major product of the organic chemical industry and many of our most familiar and useful plastics are polymers of alkenes.

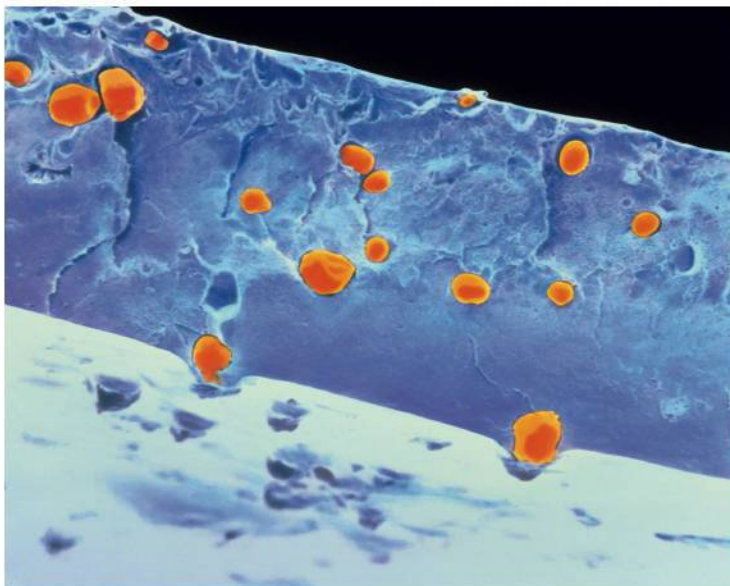
Figure 10.7 People cannot form a chain until they unfold their arms to release their hands. Similarly, alkenes must break their double bonds in order to join together to form the polymer.



As polymers are such large molecules, their structures are shown using a **repeating unit**, which has open bonds at each end. It is put in a bracket with n as subscript to denote the number of repeating units. For example, ethene polymerizes to form poly(ethene), commonly known as **polythene**.

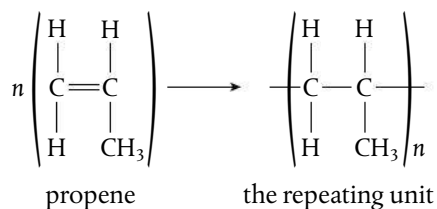


The disposal of plastics is a major global problem. The very features that make plastics so useful, such as their impermeability to water and low reactivity, mean they are often non-biodegradable and so remain in landfill sites for indefinite periods of time. It is estimated that about 10% of plastics produced end up in the ocean, causing widespread hazards to marine life. Measures to try to address this problem include developments of more efficient recycling processes, biodegradable plastics, and plastic-feeding microorganisms. A reduction in the quantities of plastic produced and used is also urgently needed – which is something for which every individual can share responsibility.

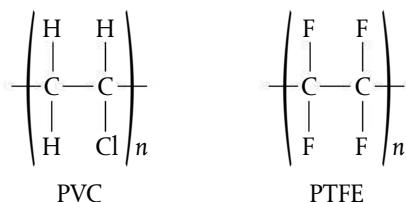


▲ Coloured scanning electron micrograph of a section through a sheet of a biodegradable plastic. The orange spheres are granules of starch embedded in the plastic. When the plastic is buried in soil, the starch grains take up water and expand. This breaks the plastic into many small fragments, increasing the contact area with bacteria in the soil which digest the plastic. Such plastics help to address the major problem of waste plastic disposal.

Similarly, propene polymerizes to form poly(propene), often called polypropylene.



Other common addition polymers include poly(chloroethene), also known as PVC, and PTFE, poly(tetrafluoroethene), often marketed as Teflon®. Their repeating units are given below.



Polymers are discussed further in Chapter 12, Option Materials.



▲ Waste plastic on a beach in Italy. Plastic waste from oceans is washed up on shores creating problems for wildlife and unsightly scenes such as this.



The example with propene shows that it is helpful to draw the structure of the monomer with the double bond in the *middle* and the other groups off at 90°. Then it is easy to see how the monomers link together when the double bond breaks.

CHALLENGE YOURSELF

- 4 Draw the repeating unit in polystyrene, given that the formula of the monomer is $\text{C}_6\text{H}_5\text{CHCH}_2$.

Polythene has excellent insulating properties and is commonly used in household containers, carrier bags, water tanks, and piping. Polypropylene is used in the manufacture of clothing, especially thermal wear for outdoor activities. PVC is used in all forms of construction materials, packaging, electrical cable sheathing, etc. There is some controversy regarding its use, linked to health and environmental concerns, and some organizations are advocating its phase-out. PTFE has a very low surface friction so is widely used in non-stick pans. It also makes up one of the layers in waterproof, breathable fabrics such as GoreTex®. Its resistance to van der Waals' forces means that it is the only known surface to which a gecko cannot stick.



In summary, alkenes:

- readily undergo addition reactions
- are used as starting materials in the manufacture of many industrially important chemicals.

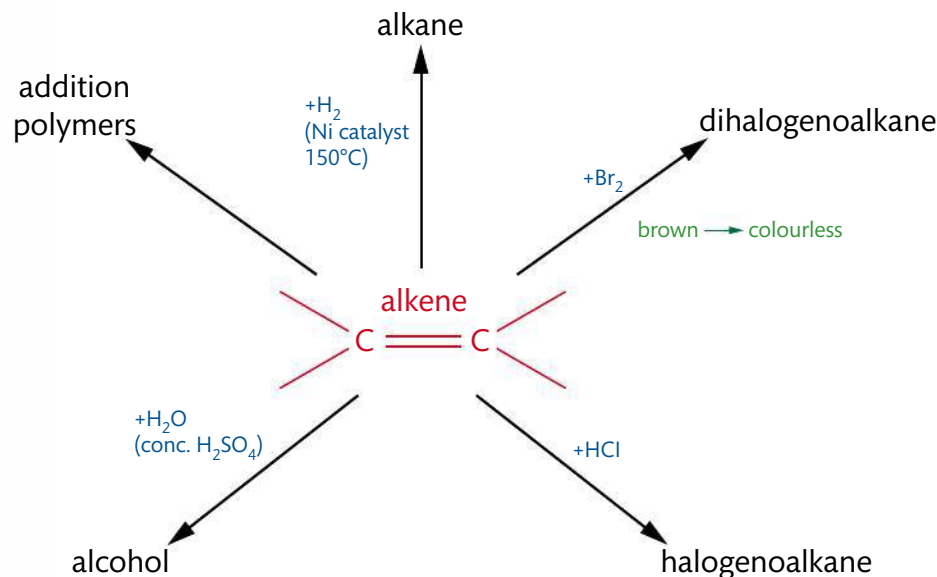


Figure 10.8 A summary of the addition reactions of alkenes.



NATURE OF SCIENCE

It is hard to assess the vastness of the scale of the impact that alkanes and alkenes have had on society. As fuels, alkanes have made energy widely available, and as chemical feedstock, alkenes have led to the synthesis of diverse plastics. Yet these innovations have come at significant cost to the environment, and the long-term consequences are still being determined. While scientific progress is responsible for both the intended and unintended consequences, it is scientists who must respond to the problems. The issues of climate change, air pollution, and plastic disposal are currently pressing areas of research and development.

Exercises

- Write equations for the following reactions:
 - incomplete combustion of pentane, forming CO(g)
 - complete combustion of butane
 - incomplete combustion of propyne, forming C(s).
- Write equations showing possible steps leading to a mixture of products in the reaction between bromine and ethane reacting together in UV light.
- Give the name and structure of the products of the following reactions:
 - $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2 + \text{H}_2$ over Ni catalyst
 - $\text{CH}_3\text{CH}=\text{CHCH}_3 + \text{conc. H}_2\text{SO}_4$
 - $\text{CH}_3\text{CH}=\text{CHCH}_3 + \text{HBr}$
- What would you observe during the following experiments carried out in test tubes at room temperature?
 - some bromine water is added to cyclohexane in a test tube covered in aluminium foil
 - a few drops of methylbenzene are burned on a watch glass
 - some bromine water is added to propane in a test tube in UV light.

Alcohols

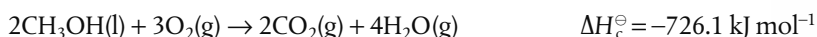
- General formula $C_nH_{2n+1}OH$
- Alcohols have $—OH$ functional group

The $—OH$ group is polar, and so increases the solubility in water of the molecules, relative to alkanes of comparable molar mass. The most common alcohol, ethanol (C_2H_5OH), is soluble in water in all proportions, as we know from its presence in alcoholic drinks.

Combustion: alcohols as fuels

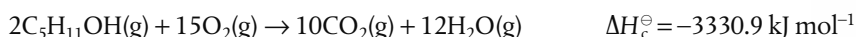
Like the hydrocarbons, alcohols burn in oxygen to form carbon dioxide and water, along with the release of significant amounts of energy. Alcohols are an important fuel, used in alcohol burners, etc. The amount of energy released per mole of alcohol increases as we go up the homologous series, chiefly due to the increasing number of carbon dioxide molecules produced.

For example, the burning of methanol:



$\Rightarrow 1:1$ ratio of alcohol : CO_2

By comparison, the burning of pentanol:



$\Rightarrow 1:5$ ratio of alcohol : CO_2 and a corresponding increase in the molar enthalpy of combustion.

As is the case with the hydrocarbons, in the presence of a limited supply of oxygen, alcohols will produce carbon monoxide instead of carbon dioxide.

Oxidation of alcohols

Combustion involves the *complete* oxidation of the alcohol molecules, but it is also possible for them to react with oxidizing agents which selectively oxidize the carbon atom attached to the $—OH$ group, keeping the carbon skeleton of the molecule intact. In this way, alcohols can be oxidized into other important organic compounds. The products of these reactions are determined by whether the alcohol is primary, secondary, or tertiary, as described on page 477.

Various oxidizing agents can be used for these reactions, but commonly the laboratory process uses acidified potassium dichromate(VI). This is a bright orange solution owing to the presence of $Cr(VI)$, which is reduced to green $Cr(III)$ as the alcohol is oxidized on heating. When writing these reactions it is often easier to show the oxidizing agent simply as '+ [O]'. The oxidation reactions of the different alcohols are as follows.

Primary alcohols are oxidized in a two-step reaction, first forming the **aldehyde**, which under prolonged conditions is oxidized further to the **carboxylic acid**. For example:



Alcohol burning in a beaker.

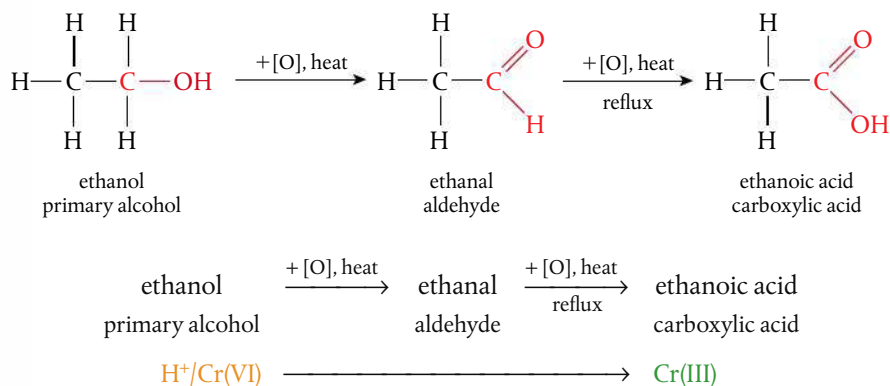


Methanol is considered to be a potential candidate to replace fuels based on crude oil. It can be burned directly as a fuel, or used in the production of hydrogen for fuel cells. It is synthesized on a large scale by the reduction of carbon dioxide and carbon monoxide.



Alcoholic drinks are widely consumed globally. However, as individuals' blood alcohol levels rise, they become impaired with loss of balance and judgment, and so most countries have set legal limits for consumption for activities such as driving. Analysis of ethanol concentration is based on samples of breath, blood, or urine. Common techniques use the oxidation reactions of ethanol in fuel cells or photocells (in which orange $Cr(VI)$ is reduced to green $Cr(III)$) or infrared spectroscopy. Alcohol abuse is a major contributor to social and economic problems in some societies.

Distillation apparatus is used to separate liquids with different boiling points. In the oxidation of primary alcohols, the aldehyde has a lower boiling point than the alcohol or the acid, and so is collected as a gas and passes into the condensing tube, which is surrounded by cold flowing water. The gas condenses back into a liquid, which is collected in the beaker at the bottom. This prevents further oxidation of the aldehyde to the carboxylic acid.



This is why when we leave a bottle of wine exposed to the air, bacteria slowly oxidize the ethanol to ethanoic acid, giving the smell of vinegar.

It is possible to stop the reaction after the first step and obtain the aldehyde product by using **distillation** to remove it from the reaction mixture. This is possible because aldehydes have lower boiling points than either alcohols or carboxylic acids, because they do not have hydrogen bonding between their molecules.

If, on the other hand, we want to obtain the carboxylic acid as the product, then we must leave the aldehyde in contact with the oxidizing agent for a prolonged period of time. This will be achieved most efficiently using apparatus for **reflux**.

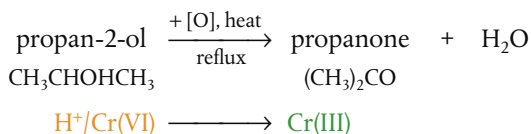
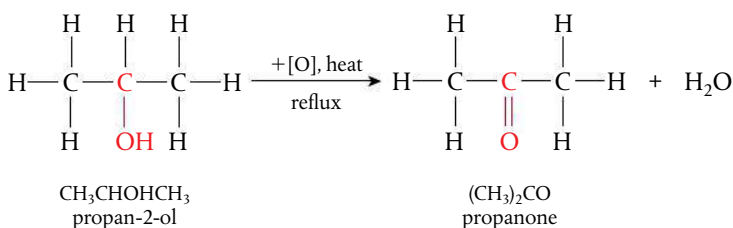
When you are asked for the conditions for the oxidation reactions, make sure that you specify acidified oxidizing agent and heat, as well as distillation or reflux.



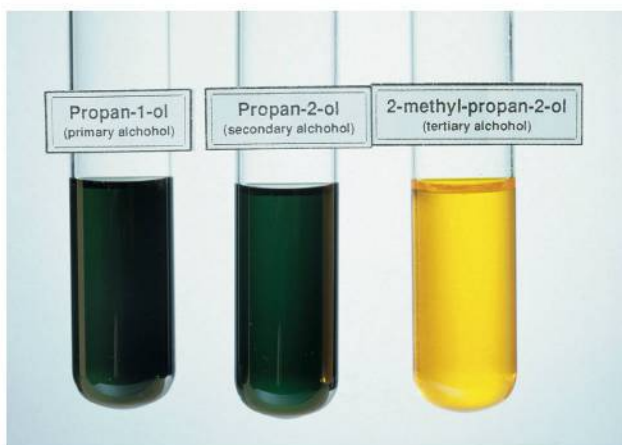
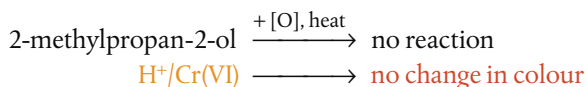
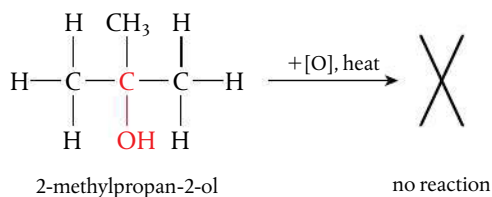
Student heating a pear-shaped flask with a Bunsen burner and using a reflux condenser. This collects and condenses vapours that would otherwise escape from the reaction mixture, and so enables the volatile components to remain in the reaction for long enough to complete their reaction. In this case it is used to oxidize a primary alcohol to completion, to the carboxylic acid.



Secondary alcohols are oxidized to the **ketone** by a similar process of oxidation.



Tertiary alcohols are not readily oxidized under comparable conditions. This would involve breaking the carbon skeleton of the molecule, which requires significantly more energy. Therefore we will not see a colour change in the potassium dichromate(VI) oxidizing agent when it is reacted with a tertiary alcohol.



The colour of the solutions after oxidation with potassium dichromate(VI) solution, $\text{K}_2\text{Cr}_2\text{O}_7$, which is orange. Primary and secondary alcohols are oxidized, to carboxylic acids and ketones respectively, and the Cr(VI) is reduced to green Cr(III) . Tertiary alcohols are not oxidized and do not react.

Summary of oxidation of alcohols

	Oxidation product	Colour change with acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$
primary alcohol	aldehyde $\rightarrow$ carboxylic acid	orange $\rightarrow$ green
secondary alcohol	ketone	orange $\rightarrow$ green
tertiary alcohol	not oxidized	no colour change



Reactions of alcohols

Full details of how to carry out this experiment with a worksheet are available online.

Exercises

- 23** State the reactants used in converting benzene to nitrobenzene. Explain how they enable the nitration reaction to occur.
- 24** Explain how the following reduction reactions are carried out in the laboratory:
- propanoic acid to propanol
 - nitrobenzene to phenylamine
 - ethanal to ethanol

20.2 Synthetic routes

Understandings:

- The synthesis of an organic compound stems from a readily available starting material via a series of discrete steps. Functional group interconversions are the basis of such synthetic routes.
- Retro-synthesis of organic compounds.

Guidance

Conversions with more than four stages will not be assessed in synthetic routes.

Applications and skills:

- Deduction of multi-step synthetic routes given starting reagents and the product(s).

Guidance

Reaction types can cover any of the reactions covered in Topic 10 and Section 20.1.

Organic synthesis is a major part of drug design. Often the process starts with a product isolated from a natural source, which acts as the 'lead' for analysis of its desirable properties, and for synthesis of related molecules. Examples include aspirin, Taxol, and mescaline. Some of these are discussed further in Option D, Medicinal Chemistry.



We can now summarize some of the reactions we have studied in this chapter, and see how they are inter-related. Separate summaries are given for aliphatic and aromatic compounds.

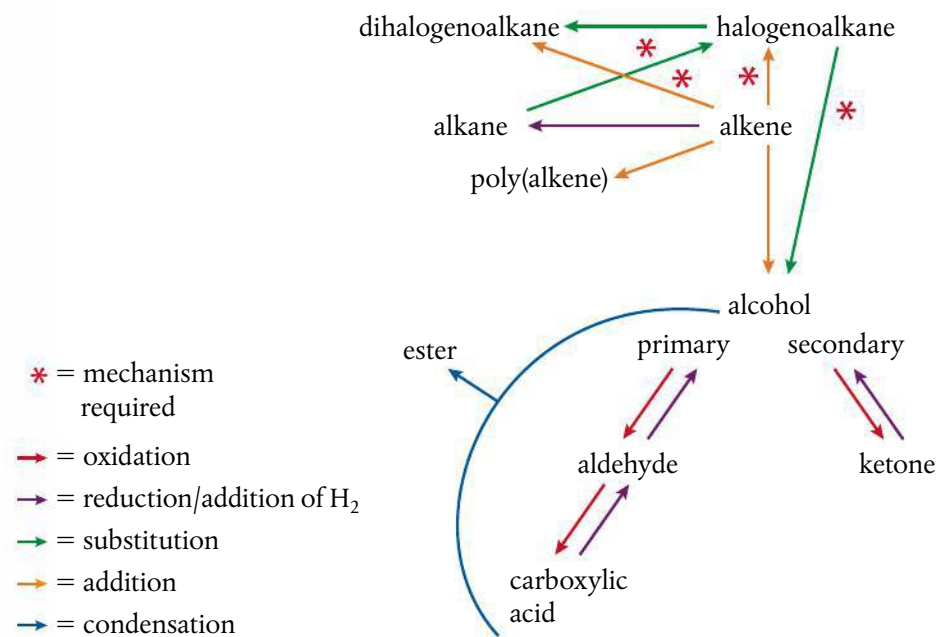


Figure 10.14 Some pathways of conversion of organic compounds.

The development of new organic compounds – from drugs to dyes, clothing to construction materials – represents a major part of modern industrial chemistry. The oil industry is the main source of organic compounds for these processes, but it

does not yield the required proportion of desired compounds. Therefore organic chemists typically have to convert compounds from one form into another, usually working with a readily available starting material and organizing several reactions in sequence, so that the product of one reaction is the reactant of the next. The series of discrete steps involved is known as a **synthetic route**. Functional group interconversions are the basis of a synthetic route, and we will look at examples from the reactions we have studied.

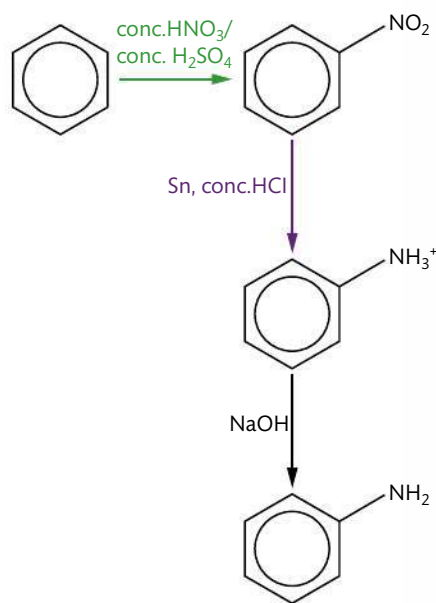


Figure 10.15 Some pathways of conversion of organic compounds.

Retro-synthesis: working backwards

This branch of organic chemistry became established as a systematic discipline largely on the basis of the work of the American chemist E.J. Corey, for which he was awarded the Nobel Prize for Chemistry in 1990. He recognized that the desired product, known as the **target molecule**, itself could serve as the starting point for our thinking. By studying its functional groups, it can be strategically broken down into progressively smaller fragments known as **precursors**. Each precursor then becomes the target for further analysis, eventually yielding a familiar molecule from which the synthetic sequence can start.

target molecule $\Rightarrow$ precursor 1 $\Rightarrow$ precursor 2 $\Rightarrow$ starting material

the symbol ' $\Rightarrow$ ' means 'can be made from'

Corey introduced the term **retrosynthetic analysis** to describe this systematic backwards approach. It has led to the efficient syntheses of thousands of molecules, and now, enhanced by computer programming to select the best route, is a major aspect of the pharmaceutical and other industries.



One of the early developments from Corey's work on retro-synthesis was the development of synthetic versions of chemicals called prostaglandins. These are involved in many aspects of physiology, especially the control of blood pressure and high pressure in the eye, known as glaucoma. The first synthetic pathway for prostaglandin F2a involved 17 steps, but this has now been refined to a 7-step pathway.

Worked example

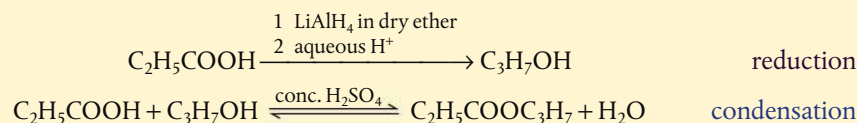
Explain how propyl propanoate can be synthesized from a single carboxylic acid. Give equations and conditions for all reactions, and state the type of reaction occurring at the functional group at each step.

Solution

Start by analysing the target molecule: the ester is the result of condensation between propanoic acid, $\text{C}_2\text{H}_5\text{COOH}$, and propanol, $\text{C}_3\text{H}_7\text{OH}$.

- Starting with $\text{C}_2\text{H}_5\text{COOH}$, a portion of it is reduced to $\text{C}_3\text{H}_7\text{OH}$ by heating with LiAlH_4 in dry ether followed by aqueous acid. The conditions must be applied for long enough to complete the reduction reaction via the aldehyde.

- Another portion of $\text{C}_2\text{H}_5\text{COOH}$ is reacted with the alcohol produced above. The mixture is heated with a catalyst of concentrated H_2SO_4 .

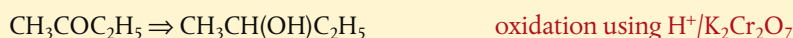


Worked example

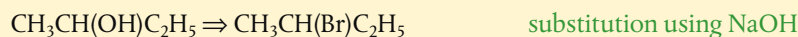
You are required to make butanone starting from any alkene of your choice. Show the steps involved in retro-synthesis to determine the identity of the alkene.

Solution

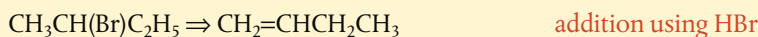
Butanone, $\text{CH}_3\text{COC}_2\text{H}_5$, is a ketone. It can be made by oxidizing a secondary alcohol.



The secondary alcohol is made by substitution of a secondary halogenoalkane.



The secondary halogenoalkane is made by addition to an alkene according to Markovnikov rules.



Therefore but-1-ene is the starting material.

(The secondary alcohol could be made directly by hydration of the alkene, but is likely to give a better yield via the halogenoalkane.)

NATURE OF SCIENCE

Corey's work on retrosynthetic analysis is an example of creative thinking and intuition in the progress of science. Using deductive reasoning and logic, he developed a set of principles and guidelines that encouraged chemists to 'think backwards' from product to starting material.



Exercises

- 25 Describe how ethyl ethanoate can be made from a single alcohol.
- 26 You are required to convert the compound 1-chlorobutane into butanoic acid. Describe the steps you would use, giving reagents, conditions, and equations for each stage.

20.3 Stereoisomerism

Understandings:

- Stereoisomers are sub-divided into two classes: conformational isomers, which interconvert by rotation about an σ bond, and configurational isomers, which interconvert only by breaking and reforming a bond.
- Configurational isomers are further sub-divided into *cis-trans* and *E/Z* isomers and optical isomers.
- *Cis-trans* isomers can occur in alkenes or cycloalkanes (or hetero-analogues) and differ in the positions of atoms (or groups) relative to a reference plane. According to IUPAC, *E/Z* isomers refer to alkenes of the form $\text{R}_1\text{R}_2\text{C}=\text{CR}_3\text{R}_4$ ($\text{R}_1 \neq \text{R}_2$, $\text{R}_3 \neq \text{R}_4$) where neither R_1 nor R_2 need be different from R_3 or R_4 .
- A chiral carbon is a carbon joined to four different atoms or groups.
- An optically active compound can rotate the plane of polarized light as it passes through a solution of the compound. Optical isomers are enantiomers.
- Enantiomers are non-superimposable mirror images of each other. Diastereomers are not mirror images of each other.

- A racemic mixture (or racemate) is a mixture of two enantiomers in equal amounts and is optically inactive.

Guidance

- The term *geometric isomers* as recommended by IUPAC is now obsolete and *cis-trans isomers* and *E/Z isomers* should be encouraged in the teaching programme.
- In the E/Z system, the group of highest Cahn-Ingold-Prelog priority attached to one of the terminal doubly bonded atoms of the alkene (i.e. **R1** or **R2**) is compared with the group of highest precedence attached to the other (i.e. **R3** or **R4**). The stereoisomer is **Z** if the groups lie on the same side of a reference plane passing through the double bond and perpendicular to the plane containing the bonds linking the groups to the double-bonded atoms; the other stereoisomer is designated as **E**.

Applications and skills:

- Construction of 3-D models (real or virtual) of a wide range of stereoisomers.
- Explanation of stereoisomerism in non-cyclic alkenes and C3 and C4 cycloalkanes.
- Comparison between the physical and chemical properties of enantiomers.
- Description and explanation of optical isomers in simple organic molecules.
- Distinction between optical isomers using a polarimeter.

Guidance

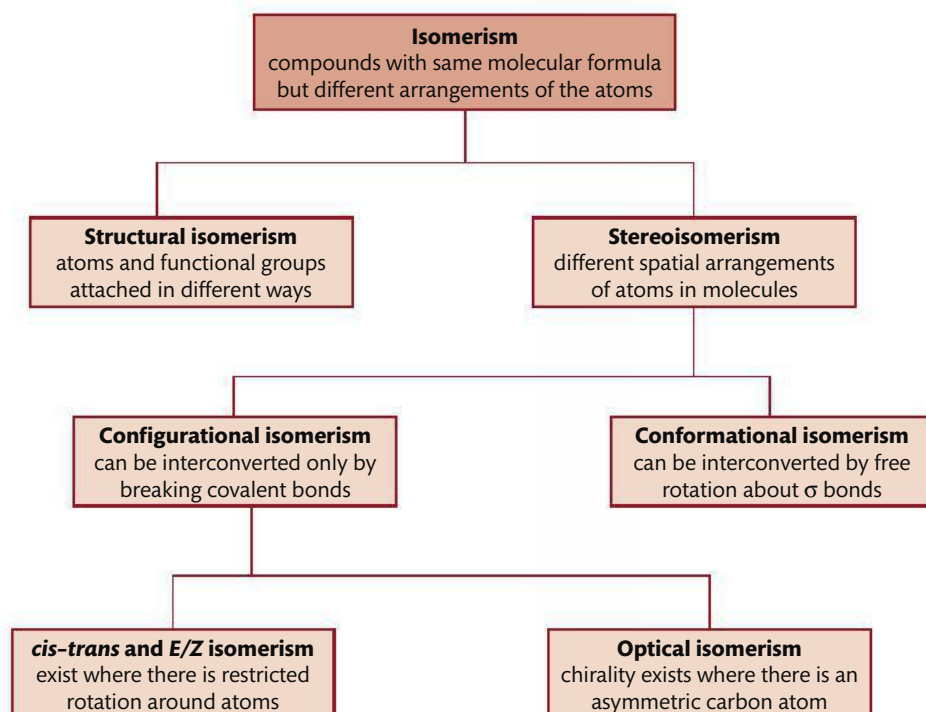
Wedge-dash type representations involving tapered bonds should be used for representations of optical isomers.

In Section 10.1 we introduced the concept of isomerism, and explored examples of structural isomers. We saw that these have atoms attached in different orders and so are fundamentally different compounds from each other. Structural isomers may have differences in the amounts of branching in their chains, or in the positions or the nature of functional groups.

Another type of isomerism, known as **stereoisomerism**, has molecules with atoms attached in the same order, but which differ from each other in their spatial or three-dimensional arrangement. Whereas structural isomers can be represented relatively easily in two dimensions on paper, the study of stereoisomerism requires three-dimensional representation.

Stereoisomerism can be further broken down into different types, as shown below.

Conformational isomers usually spontaneously interconvert by rotation, and so in most cases cannot be isolated separately. Some conformers of a compound, though, may be more stable than others, and so are favoured.



You will probably find that your understanding of this topic will be greatly helped by looking at and building three-dimensional models of the different molecules, or working with simulations of virtual molecules.

Stereoisomers differ from each other in the spatial arrangement of their atoms.

Configurational isomers have a permanent difference in their geometry. These isomers cannot be interconverted and they exist as separate compounds, with some distinct properties. We will study in turn the two main groups of configurational isomerism here.

cis-trans and *E/Z* isomers

When there is some constraint in a molecule that restricts the free rotation of substituted groups, they become fixed in space relative to each other. In these cases, the position of the groups must be described with respect to a reference plane.

There are two circumstances where these isomers arise.

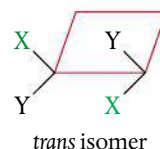
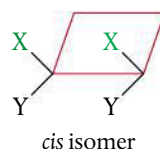
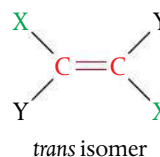
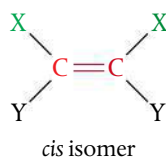
1 Double bonded molecules

The double bond consists of one sigma (σ) bond and one pi (π) bond, with the π bond forming by sideways overlap of two p orbitals. Free rotation around this is not possible as it would push the p orbitals out of position, and the π bond would break. The reference plane is perpendicular to the sigma bonds and passes through the double bond.

2 Cyclic molecules

Cycloalkanes contain a ring of carbon atoms that restricts rotation. The bond angles are strained from the tetrahedral angles in the parent alkane. For example, in cyclopropane the carbon atoms form a triangle with bond angles of 60° , and in cyclobutane the atoms form a puckered square with approximate angles of 90° . The reference plane is the plane of the ring.

When the molecule contains two or more different groups attached to the double bond or to the ring, these can be arranged to give two different isomers. The simplest examples, involving only two different substituents, form what are known as *cis* and *trans* isomers.



Cis refers to the isomer that has the same groups on the same side of the double bond or ring, while *trans* is the isomer that has the same groups on opposite sides, or across the reference plane. These prefixes are given in italics before the name of the compound.

Cis-trans isomerism plays an important role in the chemistry of vision, as light causes a photochemical transformation between the isomers of the pigment rhodopsin. *Cis-trans* isomerism is a concern in the margarine industry, where partial hydrogenation of fats leads to the production of *trans* fats, associated with some negative health effects. A widely used chemotherapy drug used in the treatment of cancer is *cis-platin*, the activity of which is dependent on its stereochemistry.